

Early Prediction of Lithium-Ion Battery Degradation with a Generative Pre-trained Transformer

Corresponding Author: Dr Yuanjian Zhang

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

The authors have tried to use the advanced deep learning techniques such as Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and Transformers for the state of health estimation of lithium-ion battery, where good accuracy can be received from the early-stage degradation.

However, the referee has not clear understand the advantages of the proposed method compared with existing methods for battery health estimation. Methods such as LSTM, CNN and Transformers have been well studies in this area for battery SOH estimation, and thus it is not clear why such a batteryGPT is needed considering the dataset is not very complex. The structure of the batteryGPT model should be clarified, and how do you train the model? And make the prediction, especially for the long-term degradation. I also suggest the authors to compare their model and accuracy with existing ones, that is, more baseline models should be added.

Thus, I can not recommend the publication of this work as it is because of the contribution and performance are not convinced.

Reviewer #2

(Remarks to the Author)

The paper presents a novel way to use transformers to predict the charging data of the entire lifecycle of lithium-ion batteries based on initial historical data. Overall, the paper is well-written and presents a compelling argument for the proposed methodology. However, there are several areas that need further clarification and expansion.

My main concern is that there are some basic concepts that, from a machine learning perspective, are simply wrong. For instance, the statement regarding BatteryGPT utilizing a Generative Pre-trained Transformer for autoregressive prediction is misleading. While the model used is GPT2, as far as I understand, it is trained from scratch therefore it does not use any pre-trained model. Moreover, the reported model size of 42.29MB lacks clarity, as model size typically refers to the number of parameters/neurons. Also, the concept of two-shot learning is not correct. One-shot/few-shot learning refers to a scenario where a model is trained with a very limited amount of labeled data, which is not the case here. Maybe a "two-stage approach" would better define it. Furthermore, there is also a lack of current literature, especially in the transformer domain for lithium-ion batteries, where there are existing works that need to be cited and discussed.

An essential aspect of the methodology is the conversion of operating features into tokens/words for model processing. However, the tokenization process requires further elucidation. Specifically, clarification is needed regarding the discretizer function, the introduction of the mentioned temporal dimension into the data, and how it coexists with the positional embedding.

The experimental design is also limited. A battery can be healthy for the first 100 cycles and degrade over the next 10 cycles and the method would predict that the battery would be healthy when in fact it is not. Other information like the IC curves could, in fact, help to unveil some silent modes that would otherwise remain invisible just by taking into account voltage, current, and temperature. Even if it works, the generalization capacities of the method should be discussed because overfitting could be the reason why the method works.

Other questions that arise:

"Predicting with the first 30% of the battery lifetime, BatteryGPT significantly outperforms baselines...". The length of the data, if 30% is used, is going to be different for each battery and Transformer models expect a fixed input size, what is used to address this?

In Figure 1 red is used to represent "bad", however, in Figure 2 it is used to represent healthy batteries (charging capacities close to its initial ones).

The positional embeddings are fixed or are they learnable?

"All input samples are divided by a window of length 256" What does this mean? maybe you meant "divided into"?

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thanks for your response. It is still not clear what is the advantage of your method. For example, many recent publications use few early cycles for the lifespan estimation. For example, 100cycles(<https://doi.org/10.1016/j.apenergy.2023.122048>), and even one cycle(<https://doi.org/10.1016/j.apenergy.2021.118134>). Why we need a complex model with 30% of cycling data?

Reviewer #2

(Remarks to the Author)

The authors have successfully addressed most of the issues raised in the previous review with clear and well-justified explanations. The manuscript now presents a more coherent argument, and the methodology is much better defined. However, I have a couple of remaining comments.

While the manuscript cites relevant works, it lacks several key references in the transformer domain for battery prediction. Studies on transformers for early knee detection and SOC/SOH estimation are missing. Including these references and distinguishing how this work differs from them would significantly improve the paper's positioning within the state of the art.

There are also some minor spelling and language issues throughout the manuscript. Although these do not detract significantly from the content, addressing them would enhance the overall readability and professionalism of the paper.

Version 2:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thanks for addressing my comments. In regarding the advantages of your method, only MIT dataset is used in your case which can not fully validate or confirm your contributions compared with existing ones. For example, the data missing case. In addition, the results have not show enough advantages compared with existing works.

(Remarks on code availability)

None.

Reviewer #2

(Remarks to the Author)

The authors have successfully addressed the concerns raised.

(Remarks on code availability)

The code looks good and can be run, although they do not provide any weights for the model, so it is not straightforward to obtain the same results as in the article.

Version 3:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

There is no more comments. Please modify figure 1 because (e) is missing in the figure.

(Remarks on code availability)

Reviewer #2

(Remarks to the Author)

The authors have successfully addressed the concerns raised.

(Remarks on code availability)

Open Access This Peer Review File is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons license, and indicate if changes were made.

In cases where reviewers are anonymous, credit should be given to 'Anonymous Referee' and the source.

The images or other third party material in this Peer Review File are included in the article's Creative Commons license, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons license and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder.

To view a copy of this license, visit <https://creativecommons.org/licenses/by/4.0/>

Reviewer #1

Thank you for your insightful feedback on our manuscript. We carefully revised the manuscript, and now we are confident that the paper's quality has been greatly improved. Your support is really appreciated. The reply to the comments is as follows:

1. The authors have tried to use the advanced deep learning techniques such as Convolutional Neural Networks (CNNs), Long Short-Term Memory networks (LSTMs), and Transformers for the state of health estimation of lithium-ion battery, where good accuracy can be received from the early-stage degradation. However, the referee has not clear understand the advantages of the proposed method compared with existing methods for battery health estimation. Methods such as LSTM, CNN and Transformers have been well studies in this area for battery SOH estimation, and thus it is not clear why such a batteryGPT is needed considering the dataset is not very complex.

Authors' response:

Thank you for your feedback. We appreciate your observations and would like to address your concerns thoroughly. Our study specifically targets the early prediction of lithium-ion battery performance degradation, which is distinct from State of Health (SOH) estimation. Here is a detailed explanation of the differences between these two tasks and the advantages of our proposed method:

1. Difference between Early Degradation Prediction and SOH Estimation:

- **SOH Estimation:** This task focuses on evaluating the current battery SOH, typically in terms of remaining capacity or overall ability to perform relative to a fresh battery [1]. SOH estimation methods often rely on current or past performance data and seek to provide an accurate snapshot of the battery's condition at a specific point in time [2].
- **Early Degradation Prediction:** In contrast, early degradation prediction aims to forecast the full lifecycle performance of a battery from its early-stage performance [3]. This involves identifying subtle initial changes that could indicate long-term degradation trends, which is a more complex task requiring the extrapolation from limited early data to predict future performance accurately [4].

2. Limitations of LSTM, CNN, and Transformers in Early Degradation Prediction:

While LSTM, CNN, and Transformers have demonstrated excellent performance in SOH estimation [2], their ability to predict full lifecycle performance based on early-stage data is limited [5]. This is primarily due to the minor initial changes in battery performance and the potential for significant performance shifts towards the end of the battery's life. These models struggle with the long-term nonlinear relationships required for accurate early degradation prediction.

3. Advantages of BatteryGPT:

- **Autoregressive Prediction Mode:** BatteryGPT employs an autoregressive prediction [6] mode, which is better suited for capturing the progressive changes in battery performance over its full lifecycle. This mode helps mitigate the shortcomings of deep learning models in fitting long-term nonlinear trends.
- **Two-Stage Prediction Approach:** Our method introduces a two-stage prediction approach. The first stage involves using a Generative Pretrained Transformer (GPT) [7] to predict charging performance across the battery's lifecycle. The second stage leverages a SOH estimator to relate the predicted charging performance to battery SOH. This approach decouples the complex early degradation prediction problem into two more manageable subproblems: autoregressive prediction of external performance changes and SOH estimation.
- **Experimental Results:** Our experimental results demonstrate that BatteryGPT significantly outperforms LSTMs, CNNs, and Transformers in the early prediction of lithium-ion battery performance degradation. This is evidenced by improved accuracy in predicting long-term performance trends from early-stage data, highlighting the efficacy of our proposed method.
- By addressing these aspects, we hope to clarify the necessity and innovation of BatteryGPT in the context of early battery performance degradation prediction. The complexity and subtleties involved in this task necessitate the advanced approach we have developed, providing a robust solution to the challenges faced by existing methods.

Thank you for considering our response. We believe that our method represents a significant advancement in the field and look forward to further discussing its implications and benefits.

- [1] Yang, S., Zhang, C., Jiang, J., Zhang, W., Zhang, L., & Wang, Y. (2021). Review on state-of-health of lithium-ion batteries: Characterizations, estimations and applications. *Journal of Cleaner Production*, 314, 128015.
- [2] Sui, X., He, S., Vilsen, S. B., Meng, J., Teodorescu, R., & Stroe, D. I. (2021). A review of non-probabilistic machine learning-based state of health estimation techniques for Lithium-ion battery. *Applied Energy*, 300, 117346.
- [3] Fei, Z., Yang, F., Tsui, K. L., Li, L., & Zhang, Z. (2021). Early prediction of battery lifetime via a machine learning based framework. *Energy*, 225, 120205.
- [4] Sulzer, V., Mohtat, P., Lee, S., Siegel, J. B., & Stefanopoulou, A. G. (2021, May). Promise and challenges of a data-driven approach for battery lifetime prognostics. In *2021 American Control Conference (ACC)* (pp. 4427-4433). IEEE.
- [5] Lin, M., You, Y., Meng, J., Wang, W., Wu, J., & Stroe, D. I. (2023). Lithium-ion battery degradation trajectory early prediction with synthetic dataset and deep learning. *Journal of Energy Chemistry*, 85, 534-546.
- [6] Lin, C. P., Cabrera, J., Yang, F., Ling, M. H., Tsui, K. L., & Bae, S. J. (2020). Battery state of health modeling and remaining useful life prediction through time series model. *Applied Energy*, 275, 115338.
- [7] Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language Models are Unsupervised Multitask Learners.

2. The structure of the batteryGPT model should be clarified, and how do you train the model? And make the prediction, especially for the long-term degradation. I also suggest the authors to compare their model and accuracy with existing ones, that is, more baseline models should be added.

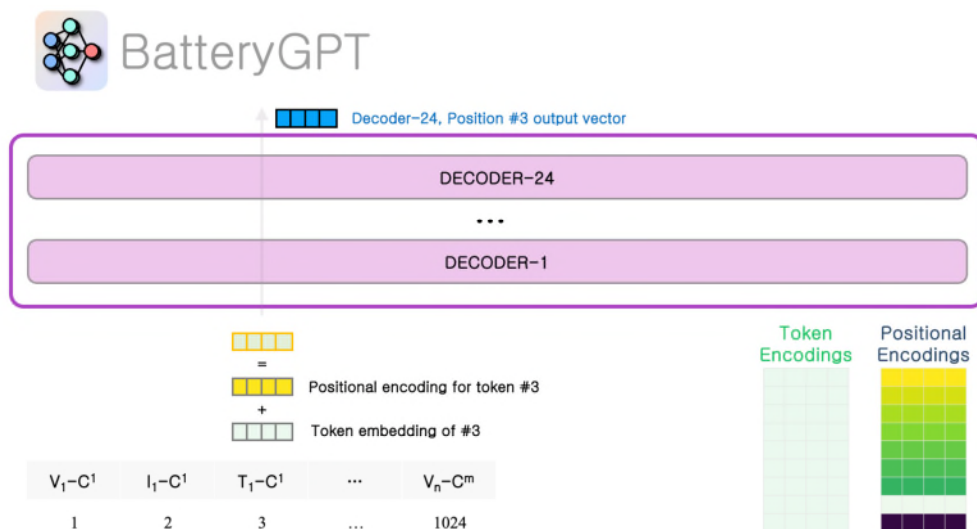
Authors' response:

Thank you for your insightful comments. We recognize the need to provide a clearer and more detailed description of the BatteryGPT model structure and training process. In response to your suggestions, we have made significant revisions and added supplementary materials to address these points comprehensively.

Revision in the Manuscript:

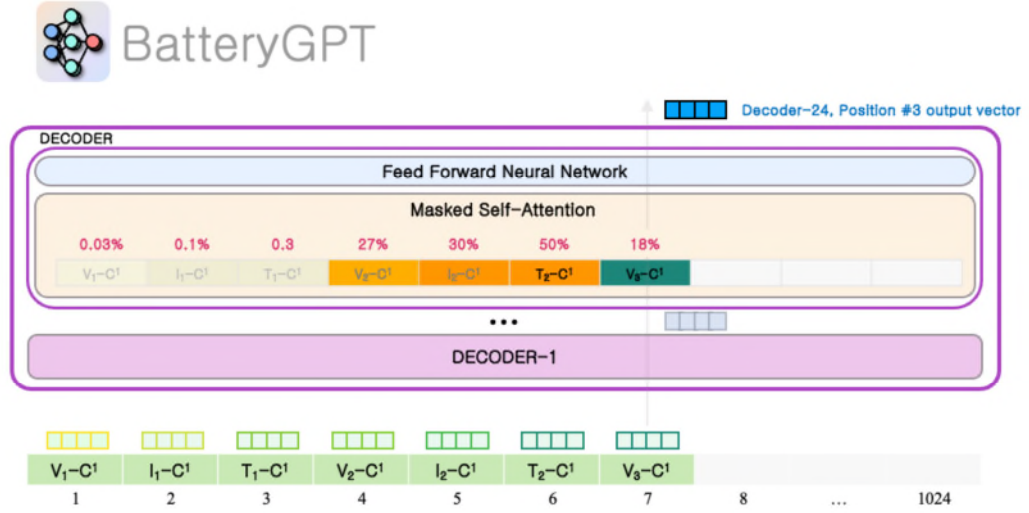
“To provide a comprehensive understanding of the BatteryGPT neural network structure, we have included a detailed model architecture diagram in the supplementary materials. This diagram illustrates the 24-layer GPT [1] used in BatteryGPT, highlighting each layer's role in the processing pipeline. Additionally, we have included an inference principle diagram (Suppl. Fig. S1) that elucidates how BatteryGPT operates from input to output, detailing the self-attention mechanisms and the autoregressive prediction process (Suppl. Fig. S4). These visual aids are accompanied by a thorough explanation of each component, ensuring that the model's functionality and architecture are transparently conveyed.”

Revision in Supplied Files (BatteryGPT Architecture and Inference Diagram):



Suppl. Fig. S1: The diagram shows the hierarchical structure of the 24-layer decoder stack, where each decoder layer processes input embeddings combined with positional encodings (Supplementary Fig. S2). The process starts with the token embedding of a voltage token (such as #1), which is then enhanced by adding the positional encoding for a specific token position (in this case, token #3). This combined input is processed sequentially through the decoders from 1 to 24, resulting in the output vector from the 24th decoder layer for the specified token

position.



Suppl. Fig. S4: This diagram illustrates the detailed mechanism of masked self-attention within the decoder layer. It shows how different tokens in the input sequence, each represented by their token embeddings, are weighted during the attention process. The percentages indicate the attention weights assigned to various tokens (e.g., V_1-C^1 , I_1-C^1 , etc.) when processing the token at position #3 (T_3-C^1). These weights reflect the importance of each token in predicting the next token in the sequence. The output of the masked self-attention mechanism is then passed through a feed-forward neural network within the decoder layer. The diagram highlights the flow from the initial input embeddings to the final output vector from decoder-24 for token position #3.

More response for the Training and Prediction Process:

We believe that there are enough details in the manuscript for the model training and prediction process of BatteryGPT, and we would like to clarify these details again:

1. Model Training:

- **Data Preparation:** We start with a comprehensive dataset containing early-stage and long-term performance data of lithium-ion batteries [2]. The data is preprocessed to normalize and structure it for model training.
- **Training Procedure:** BatteryGPT is trained using a supervised learning approach, where the model learns to predict future battery performance based on early-stage data [1]. The training involves minimizing the prediction error over the entire battery lifecycle, using a loss function that captures both short-term accuracy and long-term trend fidelity.

2. Prediction for Long-Term Degradation:

- **Autoregressive Prediction Mode:** BatteryGPT employs an autoregressive prediction mode, generating future performance data step-by-step [1]. At each step, the model uses

previously predicted data as input for the next prediction, effectively modeling the progressive degradation of the battery.

- **Two-Stage Prediction Approach:** The first stage involves the GPT-based prediction of charging performance across the battery's lifecycle. In the second stage, a SOH estimator uses the predicted charging performance to infer aging features and overall battery SOH. This two-stage approach helps to decouple the complex task into more manageable sub-tasks, enhancing prediction accuracy and robustness.

3. Baseline Models and Comparative Analysis

We selected LSTMs, CNNs, and Transformers as baselines, relying on well-established research indicating their effectiveness in SOH estimation tasks. The implementation of these models is based on the following sources:

- LSTM: Utilized for its ability to capture temporal dependencies in sequential data. [Source: You, G. W., Park, S., & Oh, D. (2017). Diagnosis of electric vehicle batteries using recurrent neural networks. *IEEE Transactions on Industrial Electronics*, 64(6), 4885-4893.]
- CNN: Leveraged for its capability to extract spatial features from structured input data. [Source: Bockrath, S., Lorentz, V., & Pruckner, M. (2023). State of health estimation of lithium-ion batteries with a temporal convolutional neural network using partial load profiles. *Applied Energy*, 329, 120307.]
- Transformer: Employed due to its powerful attention mechanism for capturing long-range dependencies. [Source: Luo, K., Zheng, H., & Shi, Z. (2023). A simple feature extraction method for estimating the whole life cycle state of health of lithium-ion batteries using transformer-based neural network. *Journal of Power Sources*, 576, 233139.]

4. Performance Evaluation:

In our experiments, we validated the performance of LSTMs, CNNs, and Transformers as backbones for early degradation prediction. We meticulously measured their performance shortcomings in fitting long-term nonlinear degradation trends.

Subsequently, we employed the same two-stage deep learning prediction framework to evaluate BatteryGPT. Our results indicate that BatteryGPT significantly outperforms these baselines in autoregressive performance prediction, demonstrating a marked improvement in both short-term accuracy and long-term trend fidelity.

By incorporating these detailed revisions and comparisons, we believe we have comprehensively addressed your concerns. Our supplementary materials now provide a transparent and thorough explanation of BatteryGPT's structure, training process, and comparative performance, underscoring the innovative aspects and superior capabilities of our proposed method.

Thank you for your valuable feedback and for considering our revised manuscript.

References:

- [1] Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language Models are Unsupervised Multitask Learners.
- [2] Severson, K. A., Attia, P. M., Jin, N., Perkins, N., Jiang, B., Yang, Z., ... & Braatz, R. D.

(2019). Data-driven prediction of battery cycle life before capacity degradation. *Nature Energy*, 4(5), 383-391.

Reviewer #2

We express our sincere gratitude to the reviewer for providing valuable comments and suggestions. After careful modification of the manuscript, we are pleased to report that the quality of the paper has significantly improved. Your support is highly appreciated. The reply to the comments is as follows:

1. The paper presents a novel way to use transformers to predict the charging data of the entire lifecycle of lithium-ion batteries based on initial historical data. Overall, the paper is well-written and presents a compelling argument for the proposed methodology. However, there are several areas that need further clarification and expansion. My main concern is that there are some basic concepts that, from a machine learning perspective, are simply wrong. For instance, the statement regarding BatteryGPT utilizing a Generative Pre-trained Transformer for autoregressive prediction is misleading. While the model used is GPT2, as far as I understand, it is trained from scratch therefore it does not use any pre-trained model. Moreover, the reported model size of 42.29MB lacks clarity, as model size typically refers to the number of parameters/neurons. Also, the concept of two-shot learning is not correct. One-shot/few-shot learning refers to a scenario where a model is trained with a very limited amount of labeled data, which is not the case here. Maybe a “two-stage approach” would better define it. Furthermore, there is also a lack of current literature, especially in the transformer domain for lithium-ion batteries, where there are existing works that need to be cited and discussed.

Authors' response:

Thank you for your thoughtful comments and for giving us the opportunity to clarify and emphasize the original contributions of our work. We have addressed each of your concerns as follows:

1. Use of "Generative Pre-trained Transformer" and Training from Scratch:

- We appreciate your attention to the details of our model "BatteryGPT." Although our model is inspired by GPT-2 [1] and trained from scratch for this specific application, we believe the term "Generative Pre-trained Transformer" remains appropriate. This terminology reflects the architectural lineage rather than the training methodology. The innovation in our work lies in adapting this architecture to a new domain where pre-training on a large corpus (long-term battery degradation in this case) is not feasible due to data scarcity.

2. Model Size and Clarity:

- We appreciate your meticulous feedback regarding the model size. This was indeed an editing error, and the correct parameter count should be 42.29M. Throughout the development and testing of BatteryGPT, we used the parameter count to measure the neural network size. We calculated the parameter count by summing the parameters across all layers of the model. Additionally, our open-source model and inference scripts (available at <https://github.com/ReparkHjc/BatteryGPT>) further confirm that the model's disk storage size is 130MB, with a parameter count of 42.29M. This clarifies that the discrepancy was a clerical error, not a conceptual misunderstanding, and does not affect the innovation and contribution of our work.
- Revisions in the Manuscript:

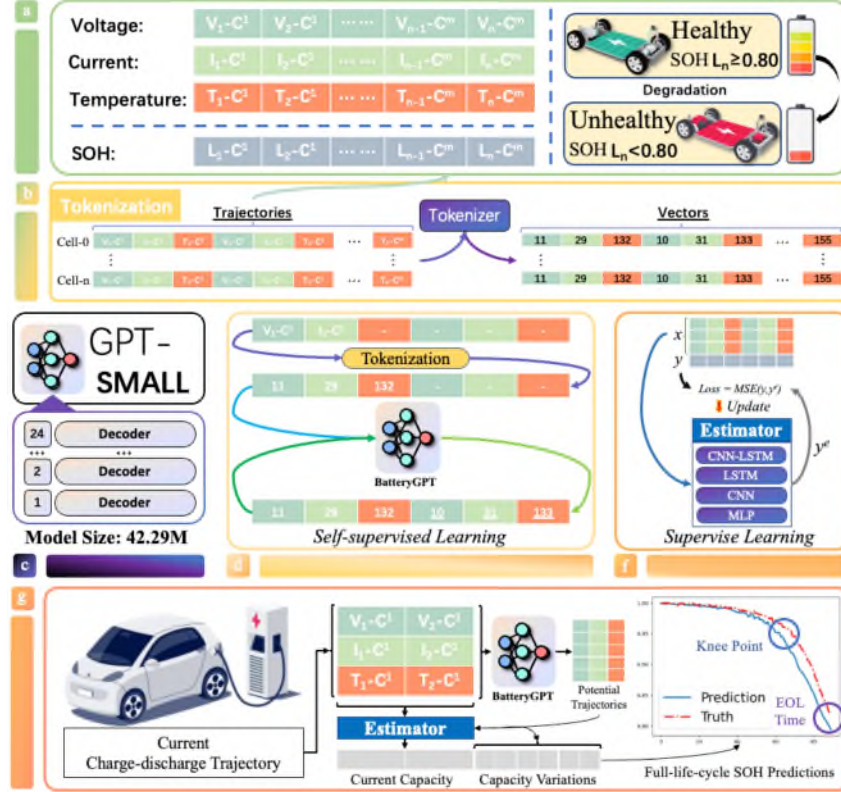


Figure 1. The diagrammatic sketch of BatteryGPT. **a** is the data collection stage, L_n denotes SOH of the n -th charging cycle. V_n , I_n , T_n denotes the voltage, current, temperature during the n -th charging cycle, with C^m denotes the m -th timestep in a charging cycle. **b** is the tokenization stage. **c** is the model structure used in this paper. **d** shows the basic workflow about the autoregressive prediction of the time-series battery features. **e** shows the estimation of SOH. **f** shows the two-stage pipeline of BatteryGPT.

“To provide a comprehensive understanding of the BatteryGPT neural network structure, we have included a detailed model architecture diagram in the supplementary materials. This diagram illustrates the 24-layer GPT [1] used in BatteryGPT, highlighting each layer's role in the processing pipeline. Additionally, we have included an inference principle diagram (Suppl. Fig. S1) that elucidates how BatteryGPT operates from input to output, detailing the self-attention mechanisms and the autoregressive prediction process (Suppl. Fig. S4). These visual aids are accompanied by a thorough explanation of each component, ensuring that the model's functionality and architecture are transparently conveyed.”

3. Misunderstanding of Two-shot Learning:

- Your concern regarding the term "Two-shot" is valuable and important. During the writing phase, we deliberated between using "Two-shot" and "Two-stage." While we acknowledge that "Two-shot" might cause confusion with concepts like zero-shot and few-shot learning, we initially chose it to reflect the application of two separate data-driven neural network models to handle the entire lifecycle of lithium

battery performance. However, we agree that "Two-stage" more accurately represents the process of performing lifecycle performance degradation prediction followed by precise SOH estimation. We have adopted "Two-stage" in the revised manuscript to avoid potential misunderstandings.

- Revisions in the Manuscript:

“In this work, BatteryGPT is developed as an early prediction method for LIB performance degradation, which is capable of accurately predicting lifecycle SOH variations, knee points and EOLs. The method involves a two-stage early prediction strategy. In first stage, a Generative Pre-trained Transformer (GPT) Small Model is employed to autoregressively predict full lifecycle charging data to effectively address the challenge of perceiving long-term nonlinear changes. In the second stage, a SOH estimator in BatteryGPT establishes a direct relationship between the predicted charging data and the future LIB SOH, assisting BatteryGPT in predicting SOH variations, knee points, and EOLs throughout the full lifecycle of LIB.”

4. Literature Review:

- We recognize the importance of situating our work within the broader context of transformer applications in lithium-ion batteries. While we have cited pioneering works, we acknowledge that we may have overlooked some recent studies directly applying transformers in this field. We have conducted a supplementary review to ensure a comprehensive discussion of relevant works. However, it is noteworthy that our preliminary literature survey suggests this specific intersection of fields is still in its nascent stages.

- Revisions in the Manuscript:

“However, the effectiveness of data-driven methods largely depends on the quality and relevance of degradation features⁷. Severson et al.⁴ extracted features from discharge voltage and capacity curves to predict battery life in early cycles, demonstrating the challenges of nonlinear performance degradation and feature analysis at the EOL for LIBs. Ma et al.²⁸ used battery capacity within a specific window as a feature to design a hybrid neural network combining convolutional neural network (CNN) and long short-term memory neural network (LSTM) to estimate capacity in the next window. Cai et al.²⁹ proposed a CEEMDAN-Transformer-DNN hybrid model for RUL prediction, suggesting that future research should use various stress features, such as temperature, voltage, and current, as input parameters to achieve more advanced battery performance degradation prediction models due to the significant influence of actual operating conditions and working environments on capacity decay behavior. Xu et al.³⁰ utilized Transformer as a global model to predict battery performance degradation, emphasizing the challenges of feature identification in early cycle data and long-term nonlinear prediction, clarifying that without additional corrections, Transformer cannot handle the severe nonlinear degradation at the EOL. Therefore, early charging data often lacks sufficient degradation features, making it difficult for machine learning and deep learning models to recognize performance changes over the entire lifespan of LIBs, limiting their accuracy in predicting long-term

degradation in the early stages³¹.”

We hope these responses adequately address your concerns and demonstrate the rationale behind our research choices. We look forward to any further comments and the opportunity to advance this discussion.

Thank you again for your invaluable feedback.

References:

[1] Radford, A., Wu, J., Child, R., Luan, D., Amodei, D., & Sutskever, I. (2019). Language Models are Unsupervised Multitask Learners.

2. An essential aspect of the methodology is the conversion of operating features into tokens/words for model processing. However, the tokenization process requires further elucidation. Specifically, clarification is needed regarding the discretizer function, the introduction of the mentioned temporal dimension into the data, and how it coexists with the positional embedding.

Authors' response:

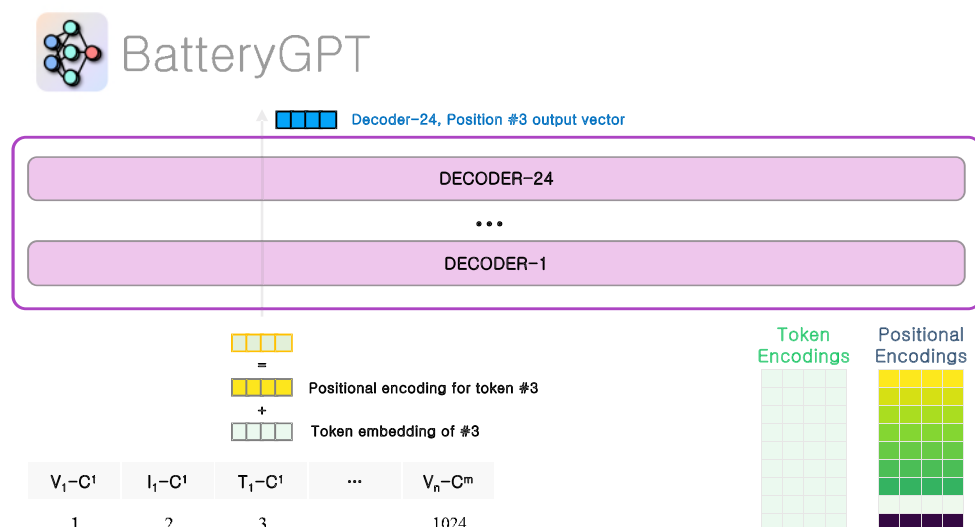
Thank you very much for meticulously reviewing our research and providing valuable suggestions. Regarding the details of the discretization process, we provide further clarification and supplementation here to ensure the clarity and completeness of the method:

1. **Detailed explanation of the discretizer function:** As you can see, we have described the tokenization methods currently supported by BatteryGPT in the Methods section of the manuscript. Each feature of battery operation (voltage V , current I , and temperature T) is discretized through the function f . Each feature is uniformly divided into specific intervals, with the discrete step size determined by the parameter η , which is set to 0.025 in our configuration. This discretization step not only compresses the data but also preserves the trend of changes, facilitating subsequent autoregressive predictions.
2. **Implementation of temporal dimension and positional embeddings:** The introduction of temporal dimension and positional embeddings in the model aims to help capture long-term dependencies and positional information in time series data. Positional embeddings are implemented by assigning a unique code to each discretized time point i , similar to the approach in the Transformer architecture. The temporal dimension is implemented through the natural ordering of the continuous input sequence, working together with positional embeddings to help the model decode the temporal and sequential structure of the input data, as shown in Suppl. Fig. S2.
3. **Enhancing relevant sections of the paper:** Based on your feedback, we have further refined the sections of the paper that describe the discretization process and model input handling. Specifically, we have added concrete examples of how the discretization step maps continuous data into a fixed discrete space and discussed how this mapping affects model performance. We hope these additions will address your concerns and further clarify our methodology. Thank you for your suggestions, and we look forward to your further feedback.

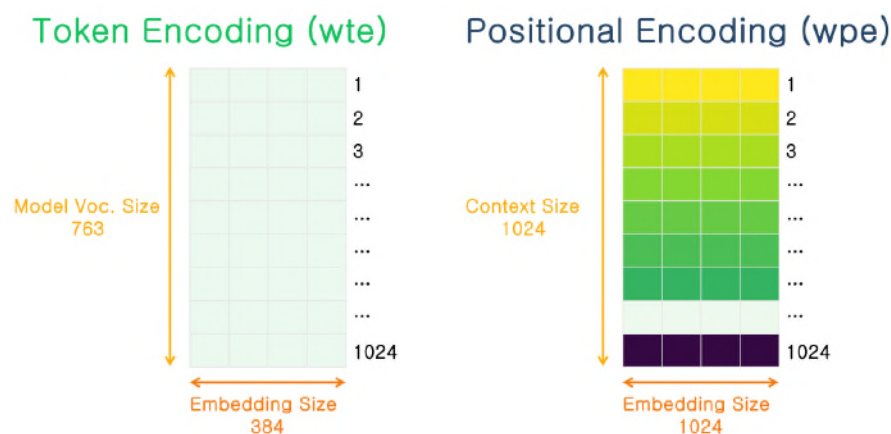
Revision in the Manuscript:

“The temporal dimension is implemented through the natural ordering of the continuous input sequence, working together with positional embedding to help the model decode the temporal and sequential structure of the input data, as shown in Supplementary Fig. S1 and Supplementary Fig. S2.”

Revision in the Supplied Files:



Suppl. Fig. S1: The diagram shows the hierarchical structure of the 24-layer decoder stack, where each decoder layer processes input embeddings combined with positional encodings (Supplementary Fig. S2). The process starts with the token embedding of a voltage token (such as V_1), which is then enhanced by adding the positional encoding for a specific token position (in this case, token #3). This combined input is processed sequentially through the decoders from 1 to 24, resulting in the output vector from the 24th decoder layer for the specified token position.



Suppl. Fig. S2: This diagram illustrates the structures of token encoding (wte) and positional encoding (wpe). The token encoding matrix (left) has dimensions of 763 (model vocabulary size) by 384 (embedding size), where each cell represents a specific token's embedding. The

positional encoding matrix (right) has dimensions of 1024 (context size) by 1024 (embedding size), with each cell encoding positional information used to maintain the order of tokens within the context.

3. The experimental design is also limited. A battery can be healthy for the first 100 cycles and degrade over the next 10 cycles and the method would predict that the battery would be healthy when in fact it is not. Other information like the IC curves could, in fact, help to unveil some silent modes that would otherwise remain invisible just by taking into account voltage, current, and temperature. Even if it works, the generalization capacities of the method should be discussed because overfitting could be the reason why the method works.

Authors' response:

Thank you for your review and valuable feedback. We have carefully considered your points regarding the limitations of our experimental design and have made the following responses and adjustments where possible:

1. Sudden Degradation of Battery Performance:

We fully agree with your mention of the abrupt performance degradation issue in batteries. This indeed poses a significant challenge for deep learning methods in predicting battery performance degradation in its early-stage. Neural networks face difficulties in predicting sharp, nonlinear changes over long timescales due to the sudden and dramatic nature of these transitions. Specifically, in the context of early degradation prediction, the battery SOH might experience a steep decline towards the end of its lifecycle. To demonstrate BatteryGPT's capability in predicting such abrupt performance changes, we included additional metrics into our experimental design. We calculated BatteryGPT's accuracy in predicting knee points and End-of-Life (EOL) points. The knee point prediction accuracy reflects the model's ability to detect the onset of rapid degradation, while the EOL prediction accuracy measures the model's effectiveness in predicting the battery's ultimate failure point. Combined with SOH prediction accuracy of BatteryGPT throughout the lifecycle, our experiments comprehensively show BatteryGPT's effectiveness in handling abrupt performance changes and providing reliable long-term predictions [1].

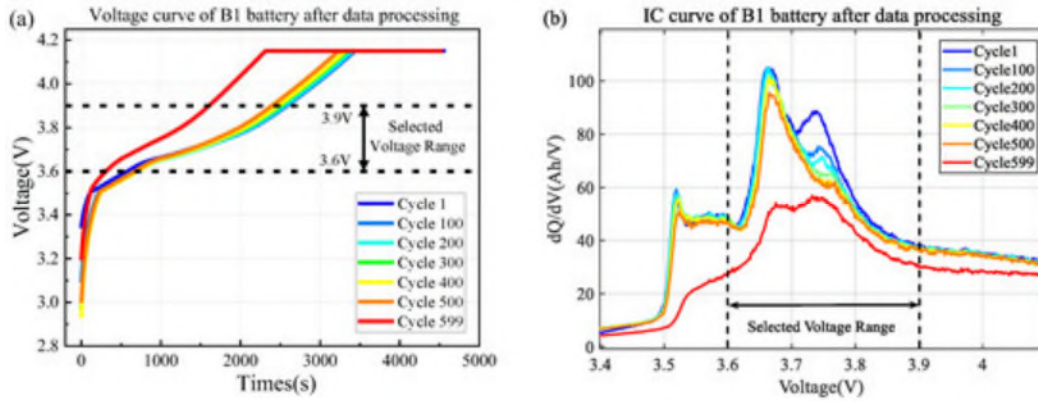
2. Inclusion of IC Curve Data:

The IC curve represents the differential of battery capacity changes against voltage changes, and by comparing the IC curves of new and used batteries, characteristics of battery aging and faults can be identified. However, in this paper, the input data used is directly observable battery performance data, and the output is an autoregressive prediction of battery performance data. Therefore, the input data lacks capacity information and cannot derive the IC curve through the differential of capacity data. Meanwhile, the objective of this study is to propose a two-stage battery early degradation prediction framework, focusing on the autoregressive prediction of battery performance data and its effective integration with the SOH estimation. Although the IC curve can reflect battery degradation, its changes are not significant in early-stage of degradation as shown in Proof Fig. 1. This lack of early change is the most critical reason it is unsuitable for early degradation prediction. [2]. In addition, we randomly selected "2018-04-12_batch8_CH33.csv" of the MIT dataset [3] to use for plotting the IC curves as shown in Proof Fig. 2, which proved to still show large changes at the end of life in the dataset used in this paper. Therefore, this paper ultimately does not use the IC curve in the

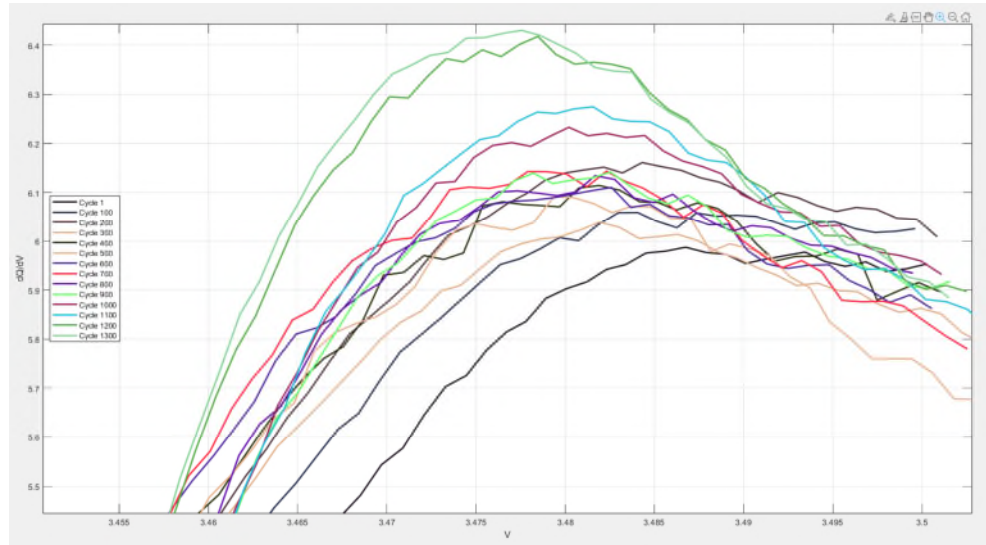
SOH estimation process. Additionally, predicting the degradation trend from more easily observable stress features during early battery cycles better aligns with practical application needs. In the revised manuscript, we highlight the importance of using stress features such as current, voltage, and temperature for early prediction of battery degradation in our literature review.

Revision in the Manuscript:

“Cai et al.²⁹ proposed a CEEMDAN-Transformer-DNN hybrid model for RUL prediction, suggesting that future research should use various stress features, such as temperature, voltage, and current, as input parameters to achieve more advanced battery performance degradation prediction models due to the significant influence of actual operating conditions and working environments on capacity decay behavior.”



Proof Fig. 1 Voltage curve and IC curve after downgrade processing of B1 cell data. (a) Voltage curve; (b) IC curve [4].



Proof Fig. 2 The IC curve of CH33 cell data.

3. Discussion on Generalization Capability:

We take your concerns about the generalization capability of deep learning models seriously. To address this, our experiments included data from 42 cells for training, covering a wide range of battery testing conditions such as different temperatures, charge/discharge rates, and cycling protocols. For testing, we used data from 4 cells under different conditions were

used to ensure a thorough evaluation of the model's generalization performance. To ensure BatteryGPT's generalization performance and avoid overfitting, we employed techniques such as k-fold cross validation, early stopping during training, and regularization. Our results show consistent performance across the test cells, demonstrating BatteryGPT's ability to generalize well to unseen data and different operating conditions.

Revision in the Manuscript:

“In addressing concerns about the generalization capability of deep learning models, this work incorporates data from 42 cells, covering a wide array of battery testing conditions such as different temperatures, charge/discharge rates, and cycling protocols. For testing, data from 4 cells under diverse conditions were employed to rigorously evaluate the model's generalization performance, with test conditions intentionally differing from those during training to assess adaptability. To ensure the generalization performance of BatteryGPT and mitigate overfitting, we employed k-fold cross-validation, early stopping, and regularization. K-fold cross-validation partitions the data into multiple folds, allowing training and validation on varied subsets, which enhances model robustness. Early stopping halts the training when the model's performance on a validation set ceases to improve, effectively preventing overfitting to the training data. Moreover, regularization techniques such as dropout and weight decay penalize model complexity, thereby reducing the risk of overfitting.”

We hope these additions and improvements address your concerns and enhance the accuracy and reliability of our research. Thank you for your suggestions, and we look forward to your further guidance.

References:

- [1] Liu, K., Tang, X., Teodorescu, R., Gao, F., & Meng, J. (2021). Future ageing trajectory prediction for lithium-ion battery considering the knee point effect. *IEEE Transactions on Energy Conversion*, 37(2), 1282-1291.
- [2] Liu, P., Wu, Y., She, C., Wang, Z., & Zhang, Z. (2022). Comparative study of incremental capacity curve determination methods for lithium-ion batteries considering the real-world situation. *IEEE Transactions on Power Electronics*, 37(10), 12563-12576.
- [3] Severson, K. A., Attia, P. M., Jin, N., Perkins, N., Jiang, B., Yang, Z., ... & Braatz, R. D. (2019). Data-driven prediction of battery cycle life before capacity degradation. *Nature Energy*, 4(5), 383-391.
- [4] Wang, J., Zhang, C., Meng, X., Zhang, L., Li, X., & Zhang, W. (2024). A Novel Feature Engineering-Based SOH Estimation Method for Lithium-Ion Battery with Downgraded Laboratory Data. *Batteries*, 10(4), 139.

4. “Predicting with the first 30% of the battery lifetime, BatteryGPT significantly outperforms baselines...”. The length of the data, if 30% is used, is going to be different for each battery and Transformer models expect a fixed input size, what is used to address this?

Authors' response:

Thank you for your meticulous review, which highlighted the need to clarify our expression regarding the implementation of the Transformer model in BatteryGPT. In fact, the input size of the Transformer model is fixed and considerably smaller than the amount of data generated

in the first 30% of the battery's lifetime. Consequently, the phrase "Predicting with the first 30% of the battery lifetime" specifically refers to utilizing data up to this point, but only a subset that fits the model's input constraints is actually used for predictions. This means that not all data from the initial 30% are employed as contextual information for BatteryGPT's predictions. Instead, BatteryGPT begins its predictions from this offset with only the data that matches the fixed input size of tokens. This selective use of data is significant because it occurs close to the 'knee point', a critical phase where the potential decline in battery performance starts to manifest. With this focused approach, BatteryGPT is able to minimize cumulative error and enhance its performance in the autoregressive prediction of battery degradation. We have incorporated this detailed explanation into the revised manuscript section that introduces the Early Prediction Start Offset (EPSO), to ensure a clearer understanding of how BatteryGPT optimally utilizes available data for predicting battery health.

Revision in the Manuscript:

“In this work, Early Prediction Start Offset (EPSO), defined as the ratio of available battery data to the full lifecycle, establishes a precise starting point for BatteryGPT's predictive analysis within the battery's lifespan. By setting different EPSOs, we quantitatively assess the performance of BatteryGPT. BatteryGPT demonstrates its leading performance in predicting LIB degradation, outperforming baselines. Even with just the first 5% of charging data (EPSO=5%), BatteryGPT significantly outperforms baseline models. It improves SOH estimation accuracy by 60.76%, knee point prediction accuracy by 31.33%, and EOL prediction accuracy by 4.23%. This trend of superior performance continues as more data becomes available. When employing the first 30% of charging cycles (EPSO=30%), BatteryGPT shows even greater improvements: SOH prediction accuracy improves by 91.80%, knee point prediction accuracy grows by 92.6%, and EOL prediction accuracy increases by 54.61%.”

5. In Figure 1 red is used to represent “bad”, however, in Figure 2 it is used to represent healthy batteries (charging capacities close to its initial ones).

Authors' response:

Thank you for your feedback. The inconsistency in chart colors you pointed out was due to our oversight. To maintain consistency and avoid confusion, we have standardized the color markings in Figures 1 and 2, with red now consistently used to indicate the "bad" state. We updated these charts in the revised manuscript.

Revision in the Manuscript:

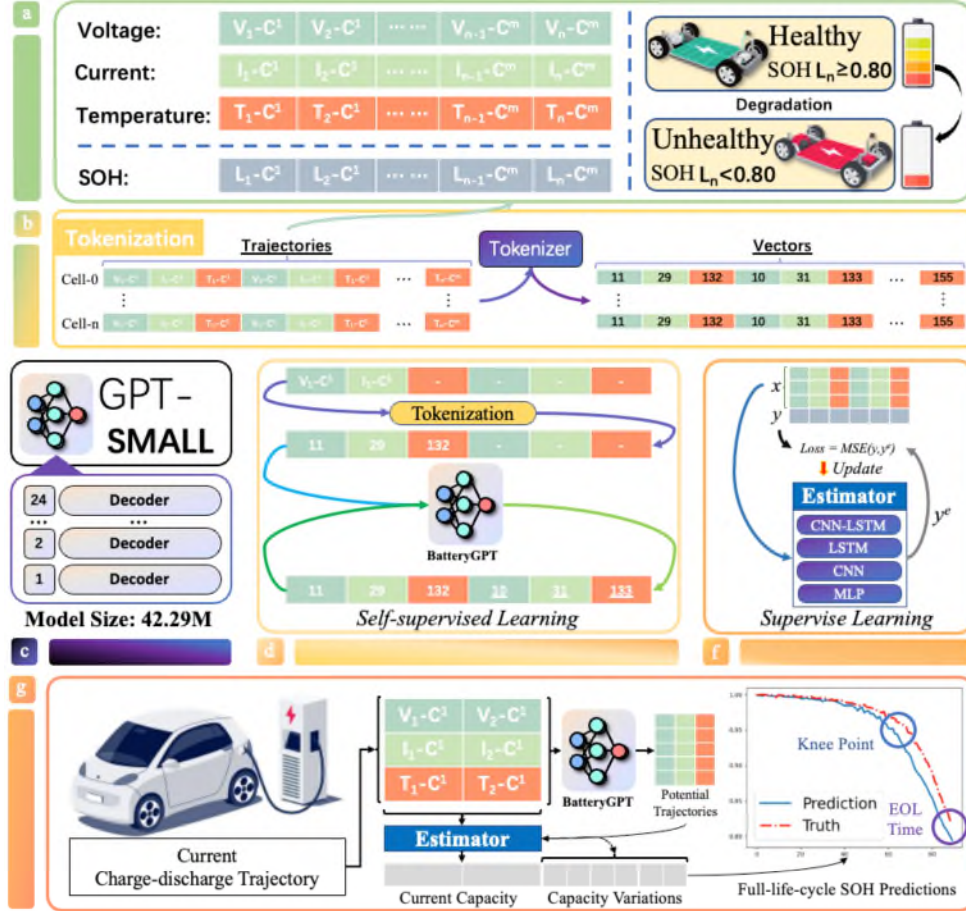


Figure 1. The diagrammatic sketch of BatteryGPT. **a** is the data collection stage, L_n denotes SOH of the n -th charging cycle. V_n , I_n , T_n denotes the voltage, current, temperature during the n -th charging cycle, with C^m denotes the m -th timestep in a charging cycle. **b** is the tokenization stage. **c** is the model structure used in this paper. **d** shows the basic workflow about the autoregressive prediction of the time-series battery features. **e** shows the estimation of SOH. **f** shows the two-stage pipeline of BatteryGPT.

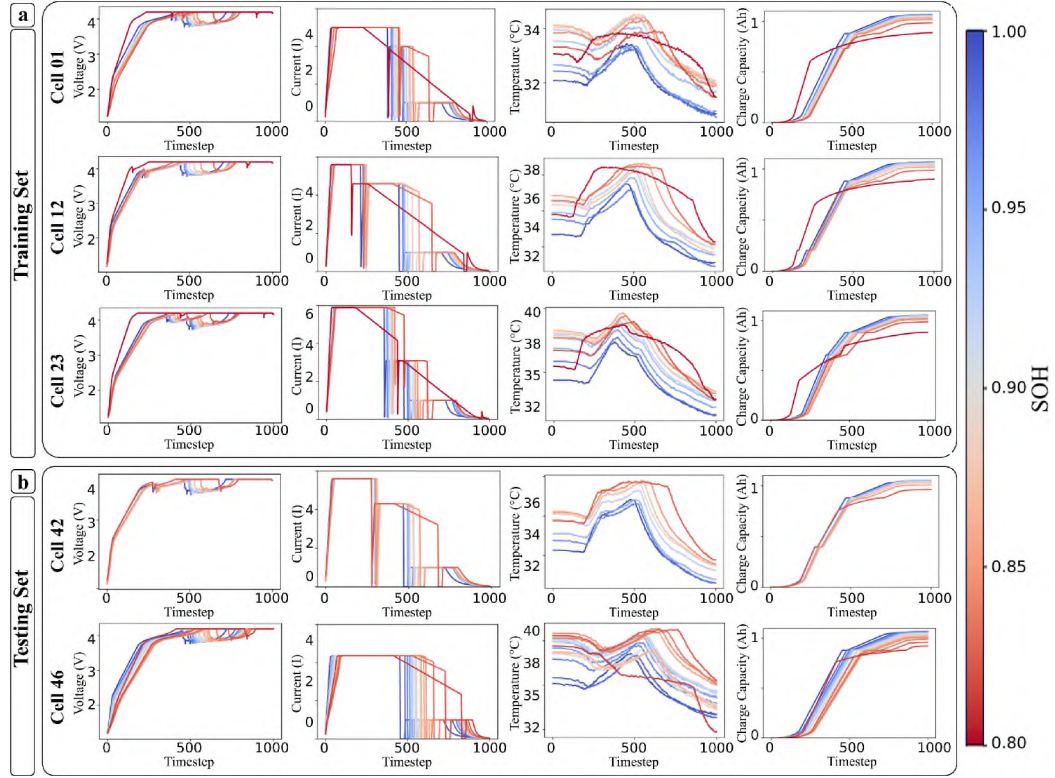


Figure 2. Overview about the battery performance degradation in the MIT dataset. Each curve represents a different charging cycle. The blue curves show that the battery charging capacities are close to its initial capacities. In contrast, the red curves indicate that the battery charging capacities are only 80% of the factory capacities, suggesting that the battery performance has significantly deterioration and reaches to the end of its lifetime.

6. The positional embeddings are fixed or are they learnable?

Authors' response:

Thank you for your insightful question regarding the positional embeddings in BatteryGPT. We appreciate the opportunity to clarify this aspect of our model.

Learnable Positional Embeddings in BatteryGPT

As shown in the Suppl. Fig. S1, the positional embeddings in BatteryGPT are indeed learnable, similar to the GPT-2 model. Here is a detailed explanation of how and why we utilize learnable positional embeddings:

1. **Positional Embeddings Mechanism:**

In BatteryGPT, we follow the same approach as GPT-2, where the positional embeddings are learnable parameters. These embeddings are added to the token embeddings to form the input to the transformer model. This combination allows the model to capture both the content and positional information of the input sequence effectively (Suppl. Fig. 1).

2. **Adaptation to Temporal Data:**

The learnable positional embeddings help BatteryGPT adapt to the temporal nature of the battery data. By learning the positional information, the model can better understand and predict the sequential patterns in the battery's lifecycle data. This learning capability enhances the

model's ability to handle the long-term dependencies and nonlinear changes in battery performance over time.

3. Flexibility and Performance:

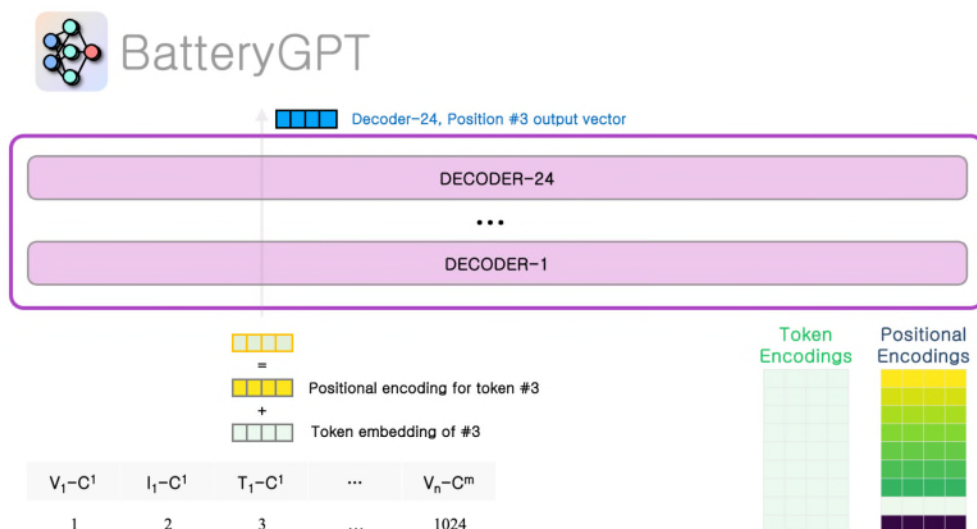
Learnable positional embeddings provide the model with the flexibility to adapt to different sequences and contexts, which is particularly useful in capturing the complex degradation patterns in battery performance data. This flexibility can lead to improved prediction accuracy and robustness, as the model can fine-tune the positional embeddings based on the specific characteristics of the training data.

4. Model Training and Implementation:

During training, BatteryGPT optimizes the positional embeddings along with the other model parameters. This joint optimization allows the model to learn the most effective way to encode positional information, improving its overall performance. As shown in the diagram, the positional encoding for each token is added to its token embedding before being processed by the transformer layers.

By employing learnable positional embeddings, BatteryGPT leverages the strengths of the GPT-2 architecture to provide accurate and reliable predictions for the entire lifecycle of lithium-ion batteries. This approach ensures that the model can effectively capture the temporal dependencies and long-term trends necessary for accurate performance prediction.

Revision in the Supplied Files:



Suppl. Fig. S1: The diagram shows the hierarchical structure of the 24-layer decoder stack, where each decoder layer processes input embeddings combined with positional encodings (Supplementary Fig. S2). The process starts with the token embedding of a voltage token (such as #1), which is then enhanced by adding the positional encoding for a specific token position (in this case, token #3). This combined input is processed sequentially through the decoders from 1 to 24, resulting in the output vector from the 24th decoder layer for the specified token position.

We hope this explanation clarifies our rationale for using learnable positional embeddings in BatteryGPT. Thank you for your valuable feedback, and we look forward to any further questions or suggestions you may have.

7. "All input samples are divided by a window of length 256" What does this mean? maybe you meant "divided into"?

Authors' response:

Thank you for your correction. You are indeed correct that the original phrasing was inaccurate. Our intention was to convey that all input samples are "divided into" windows of length 256, not "divided by" a window length. We have amend this wording to eliminate any potential misunderstandings.

Revision in the Manuscript:

“All input samples are divided into windows of length 256 with a batch size of 64. In this work, BatteryGPT are trained based on a single NVIDIA RTX4090.”

We appreciate your careful review and valuable feedback, which help us improve the clarity and precision of our manuscript.

Reviewer #1

Thank you for your insightful feedback on our manuscript. We carefully revised the manuscript, and now we are confident that the paper's quality has been greatly improved. Your support is really appreciated. The reply to the comments is as follows:

1. Thanks for your response. It is still not clear what is the advantage of your method. For example, many recent publication use few early cycles for the lifespan estimation. For example, 100cycles(<https://doi.org/10.1016/j.apenergy.2023.122048>), and even one cycle(<https://doi.org/10.1016/j.apenergy.2021.118134>). Why we need a complex model with 30% of cycling data?

Authors' response:

Thank you for your follow-up question and for highlighting the comparison with methods that utilize fewer early cycles for lifespan estimation. We appreciate the opportunity to clarify the advantages and necessity of our proposed method, BatteryGPT.

1. Research Gaps

a) Dependence on Specific Early Cycles

Existing studies on early prediction of battery degradation rely on specific early cycles (a fixed number/range of initial cycles), such as the first cycle 1141 or the in first 100 cycles 111, 51, to extract features and construct predictive models. Ning He et al. 1151 utilize the first 100 cycles to generate complex image features as inputs to a convolutional neural network (CNN). Saurabh Saxena et al. 1111 develop a data-driven method to predict the capacity decay rates and knee-point information based on the first 100 cycles. Chia-Wei Hsu et al. 1141 directly infer the remaining useful life (RUL) solely from data in the first cycle. While these methods demonstrate certain predictive accuracy in controlled experimental environments, their reliance on specific early cycles introduces significant limitations. When such data are unavailable, such as sensor failure and data loss, prediction accuracy might degrade considerably, impacting overall performance and robustness.

These studies often consider specific early cycles (e.g., the first 100 cycles 111, 51) as "representative of early-stage conditions," under the assumption that cells present insignificant performance degradation during those cycles. The data from those cycles is often employed to identify critical features of early-stage degradation in the studies. However, the assumption that "the first 100 cycles" universally represent a "stable early-stage condition" does not hold true in all practical applications. The total lifespan of a battery cell depends significantly on operational conditions such as accelerated ageing or mild usage 1161. In high-stress scenarios of accelerated ageing, which are characterized by high charge/discharge rates (e.g., 2C or higher), elevated temperatures (e.g., 45°C, 55°C, or beyond), or frequent deep discharge and fast charge cycles, the cell lifespan may shrink to merely 200 cycles or fewer. In such cases, the "first 100 cycles" constitute 50% or more of the total lifespan and can no longer be regarded as the "representative of early-stage conditions," rendering them inappropriate to represent the data bias of the early prediction. Under mild operating conditions, such as room temperature near 25°C and charge/discharge rates limited to 0.5C or 1C, battery lifespans can surpass 1,000 cycles. In such cases, the

"first 100 cycles" constitute only a small portion of the total lifespan, and the minimal degradation observed during those cycles may fail to adequately represent the long-term

performance degradation. As a result, whether under accelerated aging or mild conditions, relying solely on specific early cycles for full-lifespan prediction risks misaligning the prediction with the actual degradation.

The predictive models developed in existing studies would face difficulties in meeting the availability of the input features, if there are any data missing cases in the specific early cycles, making effective temporal extrapolation and regression analysis challenging [11, 12]. For instance, if a sensor malfunctions during initialization, preventing reliable data recording of the first cycle, the model predicated on the assumption that "the first cycle sufficiently represents the early-stage aging state of the battery" immediately lose their data availability. Since the model treat the first cycle as a critical reference point or baseline curve, the absence of this data renders them incapable of extracting key degradation features, thus hindering accurate prediction of the entire lifecycle degradation trend. It is worth noting that data-driven methods within existing studies rely on specific early cycles for training. Potential data loss or sampling bias may result in actually collected data extending beyond the temporal and numerical scope of the early-cycle data used during training, violating the independent and identically distributed (IID) assumption for data-driven methods. This triggers the Out-of-the-Distribution (OOD) issues, as shown in right of Fig. 1, where real-world sampled data deviates from the training data distribution [10]. The presence of OOD issues prevent battery prediction models that rely on specific early cycles from achieving robust generalization, significantly reducing their predictive performance.

Therefore, existing studies heavily depend on specific early cycles as the “cornerstone of extrapolation” for early prediction of battery degradation. This cornerstone collapses when critical data are missing or when distribution shifts occur. To ensure predictive accuracy and robustness in real-world production and application, it is essential to move beyond paradigms that limit model performance to specific early cycles, strengthening the capability to learn and adapt to dynamic degradation patterns across the entire lifecycle.

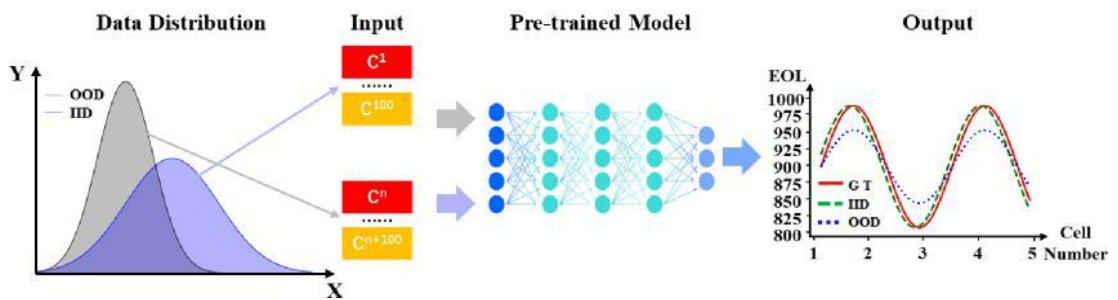


Fig. 1. Existing methods rely on a specific range of early data to study battery aging. However, the distribution of this specific range of early battery data belongs to the IID, represented by the purple region in the figure. In contrast, the distribution of battery data outside this specific range belongs to OOD, represented by the gray region in the figure. OOD refers to data that significantly deviates from the distribution of the data used during the training of a machine learning model. A model trained on a specific dataset is expected to generalize to data from the same or a similar distribution. When the model encounters data that does not belong to this expected distribution, it often struggles to make accurate

predictions or decisions, resulting in outcomes that significantly deviate from the true aging indicators.

b) Lack of Intermediate Prediction Feedback

Many deep learning or machine learning methods from existing studies for early prediction of battery degradation adopt an end-to-end or end-to-multi-end design [119, 12], directly outputting target health indicators (e.g., SOH, EOL, or RUL) from a limited set of early data, as shown in Fig. 2. Jinqiang Liu et al. [112] propose an end-to-end framework that integrates empirical degradation models with data-driven learning to estimate full-lifetime capacity trajectories based on early-life data. Joonki Hong et al. [113] develop an end-to-end deep learning framework that accelerates RUL predictions by leveraging cross-data correlations but does not assess the dynamic reliability of predictions at intermediate stages. While these methods demonstrate innovations in framework design and improvements in input and training processes, they generally lack intermediate verification capabilities, with the accuracy of prediction outcomes only confirmable after extensive cycling [114]. This paradigm of delayed validation hinders the ability to dynamically adjust early predictions and allows potential biases to accumulate, limiting insights into health state changes and early anomaly detection. If systemic bias arises in initial predictions, such errors may persist over hundreds or even thousands of subsequent cycles, causing predictions to deviate significantly from actual outcomes. For instance, in an experiment where battery lifetime is inferred from the first 100 cycles, true degradation outcomes could only be compared with predictions after completing hundreds or thousands of cycles. This validation lag renders early prediction deviations difficult to detect and quantify promptly. From the perspective of industrial production or large-scale application, this limitation significantly affects evaluations of battery safety, reliability, and economic efficiency [110]. For example, if high-risk nonlinear decay trends emerge early in degradation, delayed knowledge of inaccurate initial predictions can impede timely strategy adjustments (e.g., battery replacement or management optimization). To achieve precise monitoring of battery health states in practical scenarios and promptly capture abrupt changes such as knee points or RUL transitions, it is crucial to incorporate flexible hierarchical validation frameworks that iteratively assess the gap between "feature predictions" and "actual observations."

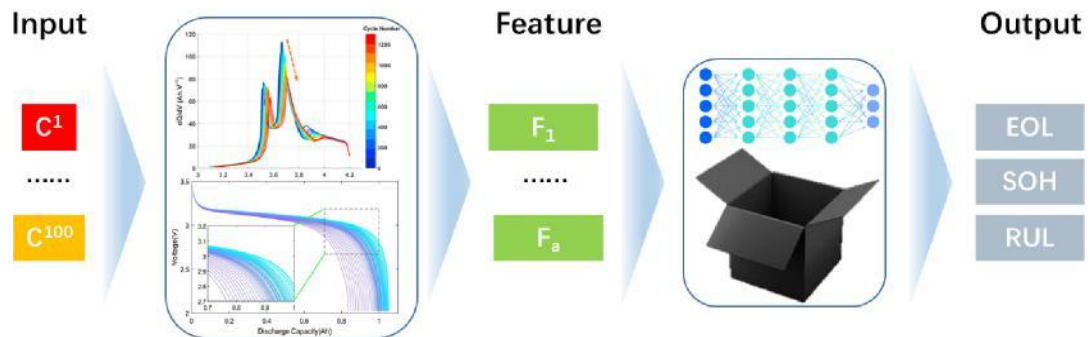


Fig. 2. The prediction pipeline of existing studies. After inputting early battery data, feature engineering is used to extract the current aging features of the battery. Subsequently, a deep learning model directly predicts the performance degradation indicators. Since the deep learning model functions as a highly nonlinear black-box model for input-output

relationships, the entire process lacks intermediate results that provide insights into the battery aging process.

c) Overreliance on Complex Feature Engineering

Existing studies substantial emphasis on multi-stage and complex feature engineering processes. Those studies derive incremental capacity curves [13] or capacity-voltage curves [14] from raw time-series data (e.g., voltage, current, temperature), followed by the application of dimensionality reduction techniques (e.g., PCA [15], t-SNE [16]) or programmatic feature selection algorithms [9] to identify the feature combinations that best indicate battery degradation patterns, as shown in Fig. 3. Ning He et al. [5] convert capacity-voltage curves, IC curves, and capacity-difference curves into three-channel images for CNN input; Saurabh Saxena et al. [1] extract and compress critical capacity decay patterns from discharge voltage-capacity curves. Afshari et al. [21] present a practical method for battery RUL early prediction by extracting 19 features from differential capacity (dQ/dV) and differential voltage (dV/dQ) curves and achieving fair accuracy compared to lasso and elastic net models. Although these methods demonstrate certain levels of success under controlled conditions, they often involve substantial implementation costs and encounter difficulties in broader applicability.

Complex feature engineering generally requires high-dimensional raw data and multiple algorithmic iterations, with each step relying on expert knowledge or in-depth understanding of experimental conditions [9]. Feature design and selection often involve iterative adjustments, such as deciding which curve to use or modifying dimensionality reduction and selection methods to emphasize subtle differences in battery aging processes. While this approach enhances prediction accuracy under specific conditions, it lacks generalizability. A shift to different battery types or varying operational environments (e.g., charge/discharge rates, temperature ranges, or sampling frequencies) often demands substantial reconstruction or re-optimization of the original feature engineering strategies. In addition, the manual feature engineering paradigm [17] lacks automated extensibility mechanisms. Those studies rely on labour-intensive trial-and-error adjustments to optimize feature extraction and dimensionality reduction algorithms, complicating the direct transfer of these processes to new scenarios. For example, in high-stress environments or extreme temperature conditions, prior knowledge may be required to hypothesize which curve segments are most likely to exhibit degradation characteristics. These segments are then integrated with feature selection and model training processes to maintain predictive accuracy while avoiding overfitting.

In this context, the applicability of the methods with multi-stage and complex feature engineering is inherently limited, which can only maintain stable prediction performance under specific types or fixed operational conditions. When attempting to generalize across different battery materials, temperature ranges, or rate configurations, previously fine-tuned feature engineering logic often fails, requiring new feature combinations to achieve the desired accuracy. In addition to lacking generalizability, frequent algorithm iterations and expert intervention require significant investment in manual tuning and experimental exploration. Differences in experimental conditions, hardware equipment, or measurement precision further hinder the direct transfer of previous feature engineering strategies. These

limitations pose substantial challenges for large-scale deployment, cross-platform application, and multi-chemistry battery lifetime prediction. To achieve efficient and universal early prediction of battery degradation, more flexible model structures and data-driven strategies must be explored to reduce dependency on manual preprocessing and expert experience, avoiding the need to rebuild the entire feature extraction process simply due to a change in scenario or data acquisition mode.

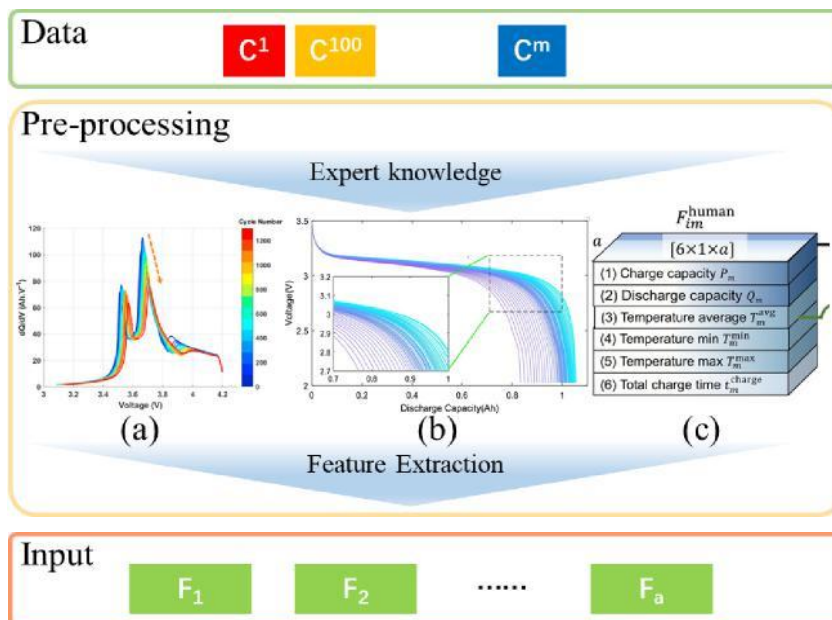


Fig. 3. The pre-processing methods for existing studies. Battery data is processed using domain knowledge from the field of electrochemistry to extract various battery aging features as inputs for deep learning methods. (a) IC curve: This represents the derivative of the capacity-voltage curve, reflecting the rate of capacity change across different voltage intervals. Based on extensive experiments, electrochemistry experts have identified that the peak positions and widths of the IC curve can serve as reliable indicators for evaluating State of Health (SOH) [7]. (b) Q(V) curve: This is the capacity-voltage curve. Through comparative performance analysis, specific voltage segments of the Q(V) curve were selected as SOH evaluation indicators [5]. (c) Human-selected features: These include statistical metrics such as maximum temperature, minimum temperature, and total charging time [4]. All these pre-processing methods are heavily reliant on domain knowledge from the field of electrochemistry.

2. How BatteryGPT Addresses These Research Gaps

Eliminating Dependence on Specific Early Charging Cycles:

The core innovation of BatteryGPT lies in overcoming the limitations of time-series-based regression analysis in existing research by eliminating dependence on specific early charging cycles. Its unique autoregressive prediction paradigm fundamentally enhances the adaptability and robustness of predictive models. Existing studies often rely on specific charging cycle data, such as the first 100 cycles or even the first cycle, to extract features and construct models. However, those studies fail when early data is missing under certain conditions, significantly limiting their robustness and generalizability of these.

BatteryGPT introduces an autoregressive prediction mechanism [18] that dynamically utilizes data from a specific context length [19] to generate predictions, rather than being restricted to specific cycles (i.e., first 100 cycles). Specifically, as illustrated in Fig. 4, the GPT-Small model in BatteryGPT leverages autoregressive capabilities to dynamically extract sequence features from recent contextual features to predict the future charging data, cycle by cycle. This design enables the model to make predictions at any charging cycle without predefined input intervals. Such flexibility significantly broadens the applicability of the model, allowing it to maintain consistent predictive performance under varying experimental conditions, charge/discharge rates, temperature environments, and battery types.

The autoregressive paradigm of BatteryGPT does not rely on existing time-series assumptions but instead achieves precise modelling of dynamic changes in battery health states through contextual learning and recursive iteration as shown in Fig. 4. This approach not only enhances robustness under conditions of missing data or non-standardized collection but also mitigates the limitations of existing research that is dependent on specific early cycles.

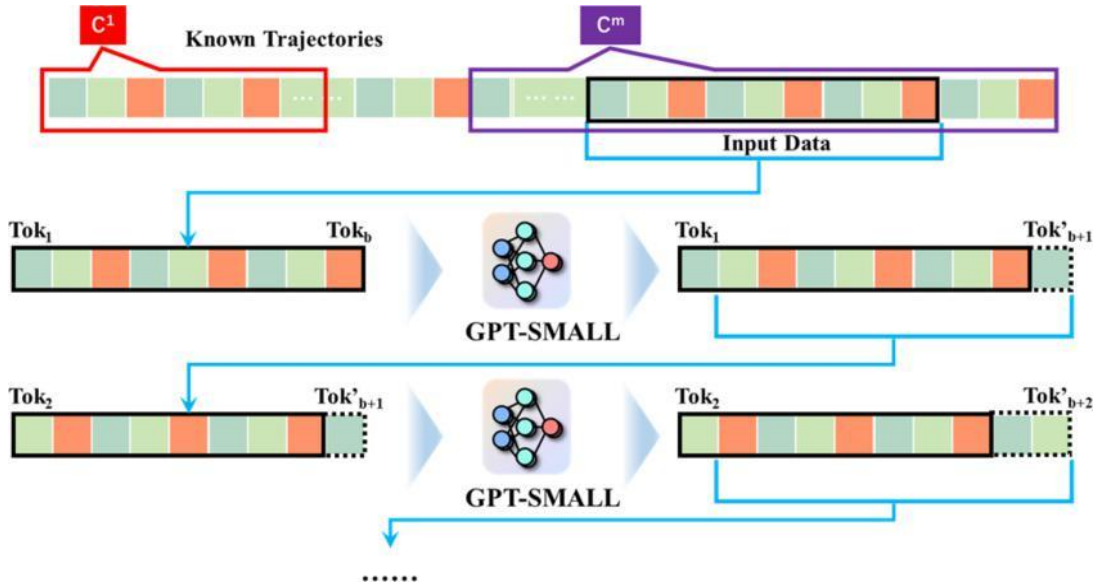


Fig. 4. The prediction process in the first stage of BatteryGPT. The input to the model consists of a fixed-length sequence of tokens (as shown in Fig. 1 in the manuscript, the detailed explanation of the physical implication of token is provided in Section Methods) from the Known Trajectories (implicating voltage, current and temperature), represented by the portion covered by the blue lines. The input data includes a smaller amount of information than a complete charge-discharge cycle. Using the GPT-Small model, an autoregressive prediction is performed, as indicated by the blue arrows. This process involves predicting the token at the next time step, appending it to the input sequence, and predicting subsequent tokens iteratively. The predicted tokens share the same physical meaning as the observed tokens. Consequently, BatteryGPT can predict the operational trajectory of the entire battery lifecycle based on the observed operational states at any given point. C^m represents the battery data for the m -th charge-discharge cycle.

b) Two-Stage Prediction Framework with Intermediate Feedback

The two-stage prediction framework of BatteryGPT offers significant advantages in addressing the "lack of intermediate prediction feedback" issue prevalent in existing studies. Existing end-to-end or end-to-multi-end architectures for battery lifetime prediction often require long charging cycles or completing the entire test to validate early predictions using terminal indicators (SOH, knee point, or EOL) [9]. This results in extended validation cycles and a lack of transparency in the prediction process. As illustrated in Fig. 5, BatteryGPT addresses these limitations by generating a complete battery operational trajectory in the first stage and combining it with a pre-trained SOH estimator in the second stage to produce key health indicators. This successfully embeds an intermediate feedback mechanism within the prediction process, enabling multi-level observation and evaluation of predictions before the final conclusions are reached.

In the first stage, the GPT-small model utilizes the current charging/discharging state and existing contextual features to autoregressively predict a full-lifecycle operation trajectory of the battery across timesteps, as is illustrated in Fig. 4. This trajectory not only encapsulates the primary dynamic features of the battery (including voltage, current and temperature) but also captures short-term correlations via specific contextual windows. Since this stage does not directly produce final health indicators, it allows for a preliminary evaluation of prediction results at the trajectory level. Researchers and engineers can assess the reasonableness of early predictions before the completion of extensive charging cycles. Compared to traditional "black-box" end-to-end methods that directly output knee point or EOL results, this approach substantially enhances the transparency and traceability of the prediction process. Specifically, in the Results section of the manuscript, we conduct a detailed analysis of the trajectory prediction accuracy in the first stage. Figures 4 and 5 in the manuscript provide a detailed comparison showing the trajectory prediction errors of the GPT-Small model relative to other advanced autoregressive solutions. Notably, the GPT-Small model exhibits commendable tracking performance for the battery charge cycle data, with its predicted feature changes remaining consistently close to the actual observations than SOTA methods, further demonstrating its reliability and accuracy in trajectory prediction tasks.

Once the trajectory prediction is complete, the pre-trained SOH estimator in the second stage uses the trajectory as an input feature sequence to generate a full-lifecycle SOH curve and corresponding terminal indicators, including SOH, EOL and knee points, as shown in Fig. 5. This approach delivers outputs equivalent to or richer than existing studies and transparently separates the prediction process into two independently observable and evaluable layers: "trajectory generation" and "health state inference." Through this layered design, BatteryGPT provides intermediate predictive information before results are output. Researchers can compare this information with real data (partially completed charging cycle records) to identify potential prediction biases or uncertainties earlier. If significant discrepancies emerge during preliminary trajectory-to-actual data comparisons, the research team can review test procedures or early data before completing long-cycle experiments, promptly verifying predictive robustness or exploring whether degradation patterns have undergone abrupt changes.

By offering observable and evaluable intermediate outcomes during the prediction process, BatteryGPT effectively mitigates the key challenge of "lack of intermediate prediction feedback" that has troubled traditional battery lifetime prediction approaches. On one hand, trajectory predictions in the first stage serve as a reference for early reliability assessments. On the other hand, the second stage's in-depth interpretation of trajectories and health state inference fully clarifies the relationship between "intermediate results" and "final outcomes." This granularity allows researchers or engineers to progressively track the rationality of the model's inference process at finer levels. These improvements not only reduce the time costs associated with validation feedback loops but also decrease over-reliance on potentially biased terminal indicators, offering greater transparency and stronger interpretability for battery health management and early-stage degradation detection.

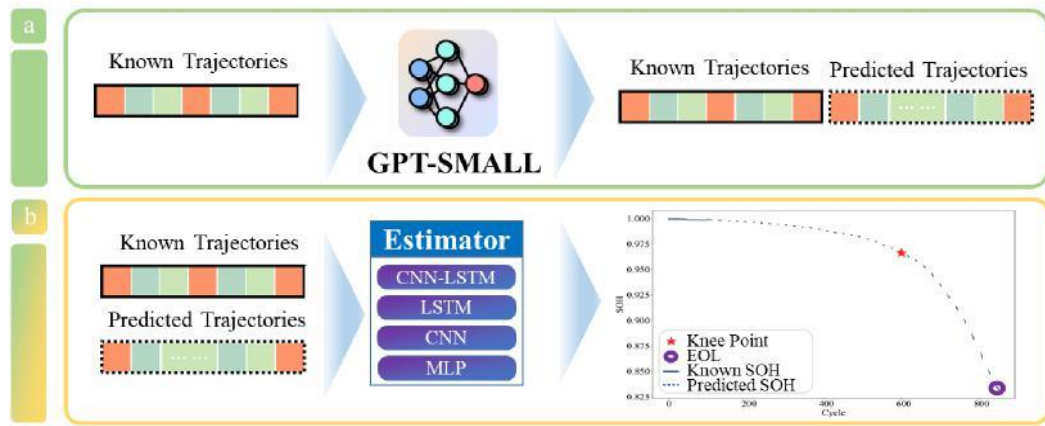


Fig. 5. The two-stage prediction framework of BatteryGPT. (a) This stage involves the autoregressive prediction process detailed in Fig. 4. The Known Trajectories are input into the GPT-SMALL model to generate the completed Predicted Trajectories for the battery's entire lifecycle. (b) In this stage, the Known Trajectories are processed by an estimator to produce the Known SOH values, while the Predicted Trajectories are processed by the same estimator to obtain the Predicted SOH values. This results in a complete SOH curve for the battery's lifecycle, enabling the estimation of the EOL and the identification of other inflection points in battery aging parameters.

c) Eliminating Complex Feature Engineering

BatteryGPT avoids the reliance on complex feature engineering, moving beyond the intricate manual feature extraction and selection processes prevalent in existing studies. Existing early prediction methods of battery degradation often require constructing complex features combinations with specific dimensionality reduction or mathematical fitting techniques to capture subtle degradation patterns, such as incremental capacity curves and capacity-voltage curves. These methods perform well under the controlled experimental conditions but face challenges with generalizability and scalability. Changes in battery types, usage environments, or sampling resolutions often require researchers to redesign feature extraction workflows or adjust feature parameters, significantly increasing deployment difficulty and costs.

BatteryGPT simplifies the dependency on manual feature engineering. Its core approach

directly utilizes raw observable data, such as discharge voltage, current, and temperature, only requiring the standardized flattening and concatenation at the model input layer without constructing highly customized features such as incremental capacity curves or capacity difference curves as shown in Fig. 6. This "minimal preprocessing, robust modelling" paradigm retains the high-dimensional information from raw time-series data within the learnable space of the end-to-end model. Its internal autoregressive modules and deep representation layers automatically capture complex degradation correlations and patterns [19].

This feature simplification does not imply a weakening of model functionality. Instead, it demonstrates stronger applicability in real-world scenarios. On one hand, key observables of battery performance already encompass critical signals of degradation. By directly modelling parameters such as discharge voltage, current, and temperature, BatteryGPT effectively captures essential elements required for most chemical systems and operating conditions. On the other hand, reducing reliance on manual feature selection and engineering workflows means less frequent rewriting or adjustment of preprocessing algorithms in different experimental settings, minimizing uncertainties and intervention costs associated with parameter selection and extending the model applicability.

BatteryGPT ensures robust segmented learning of high-dimensional raw data through its autoregressive prediction mechanism while preserving the original information structure with minimal standardized processing (e.g., flattening and concatenation) at the input stage. These design choices collectively aim to reduce the investment in manual preprocessing and embrace a "data-driven learning" paradigm instead of a "manually designed features" mindset. With the continuous accumulation of large-scale battery experimental and real-world data, this approach not only suits to accommodate heterogeneous data sources better but also provides a solid foundation for model optimization and scenario adaptation.

By reducing or eliminating reliance on manual feature engineering, BatteryGPT fundamentally transforms the workflow for early prediction of battery degradation, overcoming scalability limitations inherent in prior feature-engineering-based methods.

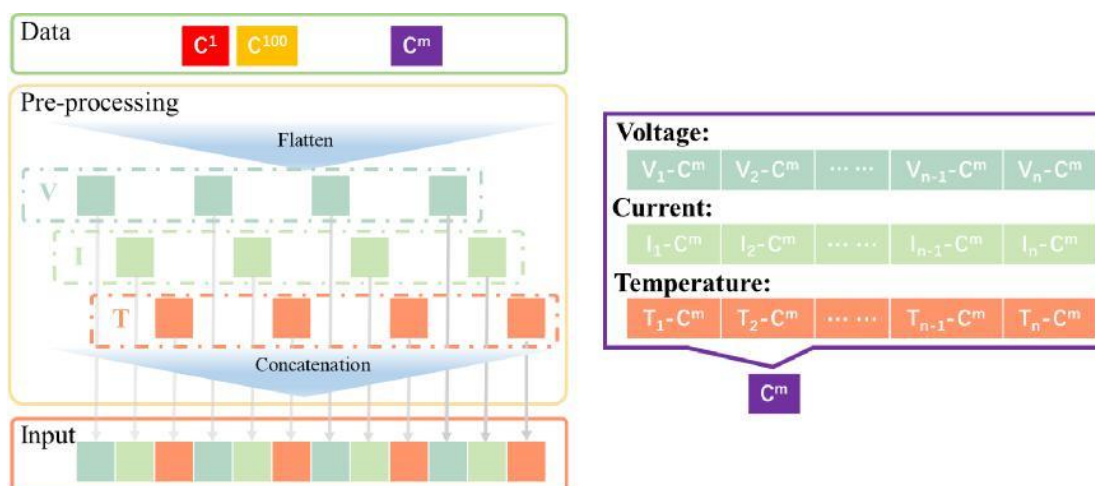


Fig. 6. The pre-processing process of BatteryGPT. Battery data requires only two spatial operations, flattening and merging, to be converted into the input sequence for the BatteryGPT network. This eliminates the need for additional expert knowledge or complex dimension reduction operations. Three-dimensional battery data is flattened into one-

dimensional data, with each position corresponding to specific battery data. For example, the first position represents voltage, the second represents current, the third represents temperature. After flattening, all charge-discharge cycle data is merged sequentially according to the cycle order, forming the format of the input data for BatteryGPT. C_m represents the battery data for the m -th charge-discharge cycle.

3. Potential Scalability and Scientific Value of BatteryGPT

BatteryGPT establishes a solid foundation for integrating advanced AI techniques, including causal inference, Transformers, and large-scale generative pre-trained models, into battery management systems. Recent advancements in large language models (LLMs) and their scaling laws indicate that increasing model capacity and training dataset size consistently improves predictive performance across domains [20]. By leveraging these developments, BatteryGPT can take advantage of continuously growing streams of battery lifecycle data, enhancing the accuracy and robustness in predicting battery health.

This progress holds great promise for accelerating innovations in battery management. By combining more comprehensive datasets with larger models, BatteryGPT has the potential to improve diagnostic fidelity, simplify maintenance processes, and ultimately extend battery lifetimes under various operating conditions. These advancements highlight the transformative potential of applying advanced AI technologies to sustainable energy storage solutions.

4. Supplementary Explanation Regarding EPSO = 30%

Due to significant variations in the total lifespan of batteries under different operating conditions (e.g., charge rates, operating temperatures, as shown in Fig. 7), this study employs the Early Prediction Start Offset (EPSO) parameter (e.g., 30%) to specify the starting point of the prediction within the degradation process. Compared to directly referencing a specific cycle (e.g., the 100th cycle), EPSO offers greater flexibility and precision in reflecting the actual degradation stage at which BatteryGPT operates under different operating conditions. Specifically, EPSO represents the percentage position in the battery lifecycle where autoregressive prediction starts, describing only the prediction's starting point (the offset from the lifecycle begin) rather than indicating that all data within the EPSO range is used as input. In this study, the early cycle stage is defined as the first 0–30% of the lifecycle (EPSO in [0%, 30%]), following a balanced segmentation approach that ensures equitable sampling across different degradation stages. For instance, at EPSO = 30%, BatteryGPT does not consume the entire 30% of lifecycle data but instead extracts a fixed-length context around the 30% offset point to perform autoregressive predictions. In other words, the model's "actual data usage" is constrained by the fixed size of the GPT input window rather than by utilizing the full 30% of lifecycle data. In this study, the input context length for BatteryGPT is fixed at 256 tokens, where each set of three tokens represents a single record of discharge voltage, current, and temperature. Consequently, 256 tokens correspond to approximately 85 data points, which is less than the data generated in one complete charge-discharge cycle [4, 9]. As shown in Fig. 7, whether EPSO is set to 5%, 30%, or any other value, BatteryGPT consistently processes the same amount of raw data points as input. This data efficiency is significantly lower than that of existing

methods that rely on the complete first 100 cycles or an entire charging cycle as feature inputs [4,5].

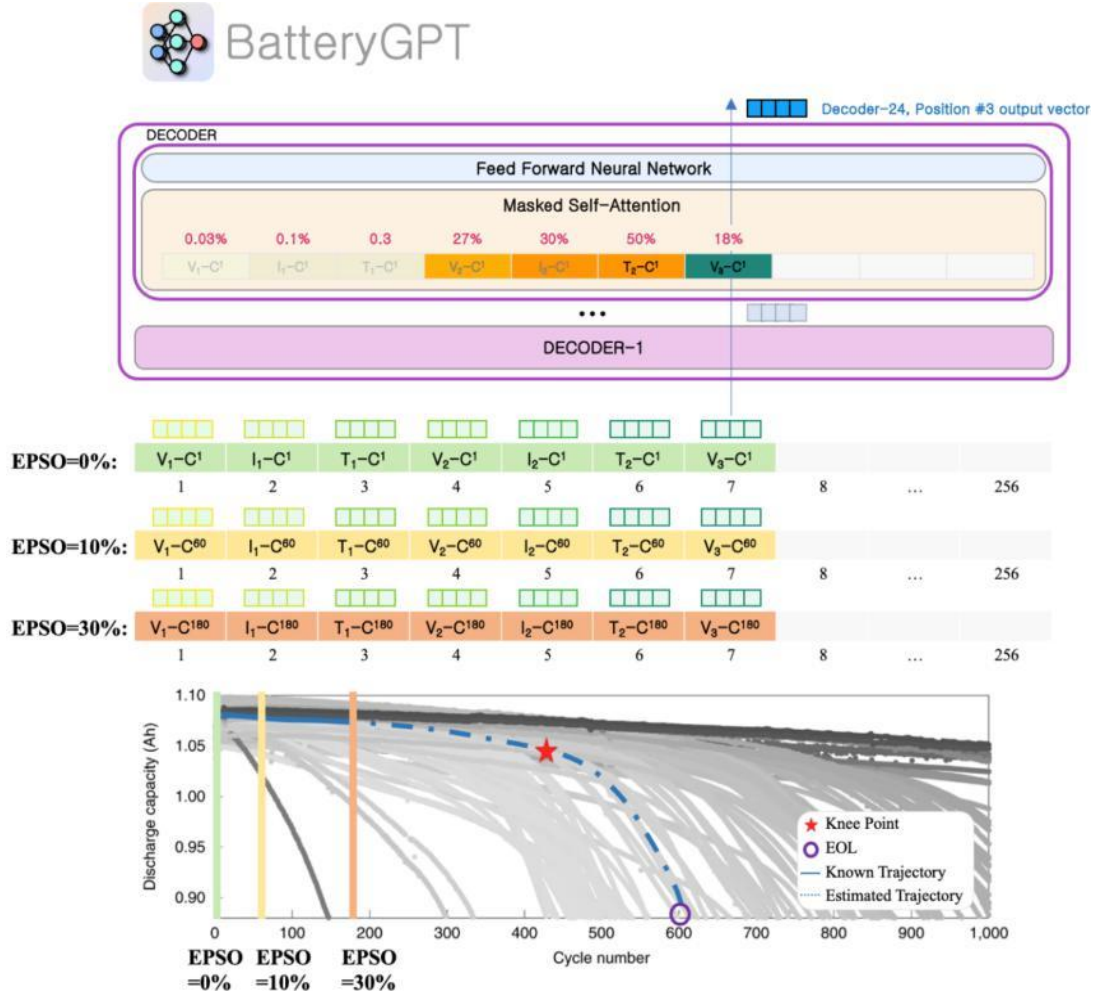


Fig. 7. Predictive starting points and data utilization of BatteryGPT under different EPSO settings. Taking a battery with a lifecycle of 600 cycles as an example. When EPSO = 0%, the input to the model consists of 256 tokens starting from the 1st cycle. When EPSO = 10%, the input to the model consists of 256 tokens starting from the 60th cycle. When EPSO = 30%, the input to the model consists of 256 tokens starting from the 180th cycle.

At the higher offset value of EPSO = 30%, the exceptional performance of BatteryGPT is not attributed to the use of more data but rather to the proximity of this offset point to the nonlinear degradation phase of the battery. This allows the model to directly leverage the progressively emerging degradation characteristics at this stage (as illustrated in Fig. 8). In the early cycles, batteries typically exhibit insignificant nonlinear degradation signals. However, as cycles accumulate, indicators such as capacity, discharge curve morphology, and early signals of knee points become increasingly evident. A higher EPSO value enables BatteryGPT to capture these critical signals from the outset of prediction, and despite the 256-token context limit, the model achieves superior accuracy and stability in modelling degradation trends and identifying knee points.

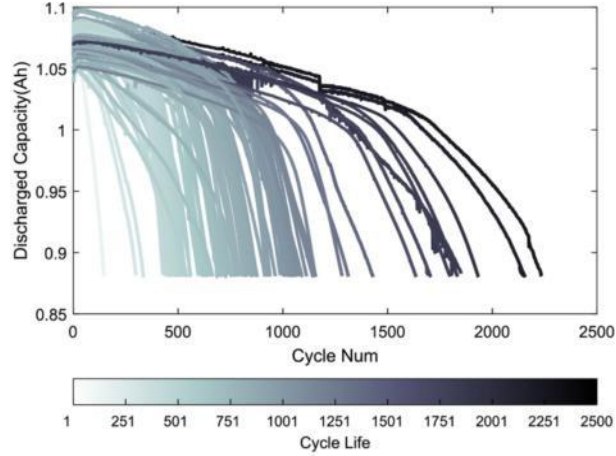


Fig. 8. The full life cycle lengths of different cells [4]. Cells can exhibit significant variations in lifespan under different cycling conditions.

In summary, while $EPSO = 30\%$ nominally represents an offset at 30% of the battery lifecycle, it does not imply that BatteryGPT uses all the data within this range. Instead, the model relies on a strictly defined 256-token context window to extract a small segment of time-series data, achieving outstanding predictive performance when degradation characteristics are more pronounced. This mechanism, balancing data efficiency with degradation dynamics, enables BatteryGPT to eliminate dependence on large volumes of early-cycle data while maintaining robust and accurate capacity degradation predictions under varying EPSO settings.

5. Summary of Revisions

We recognize that the reviewers' concerns regarding the scientific value of our innovations and potential misunderstandings about BatteryGPT's data usage stem from incomplete literature review, insufficient explanation of how research gaps are addressed, and ambiguous descriptions of EPSO in the originally submitted manuscript. In the revised manuscript, we have addressed these points as follows:

1. Enhanced Literature Review

We have expanded and refined the literature review section, detailing the key gaps in existing research, including the reliance on complex feature engineering, the lack of intermediate prediction feedback, and the dependence on specific early charging cycles. By citing multiple high-impact references, we have emphasized BatteryGPT's unique value in addressing these gaps, thereby articulating the scientific innovation of our study more clearly.

Revision in the Introduction:

“Costa et al.³¹ develop ICFormer, a Transformer-based deep learning model that leverages self-attention mechanisms to analyse incremental capacity curves for early knee detection in lithium-ion batteries, highlighting the model's ability to identify degradation patterns and predict battery health indicators more effectively than traditional methods. Zhao et al.³² propose a novel Transformer-embedded model, combining Transformer-based feature extraction with enhanced unscented Kalman filter for SOC estimation, addressing challenges in aging feature identification under varying temperature and operating

conditions. Therefore, early charging data often lacks sufficient degradation features, making it difficult for machine learning and deep learning models to recognize performance changes over the entire lifespan of LIBs, limiting their accuracy in predicting long-term degradation in the early stages³³.

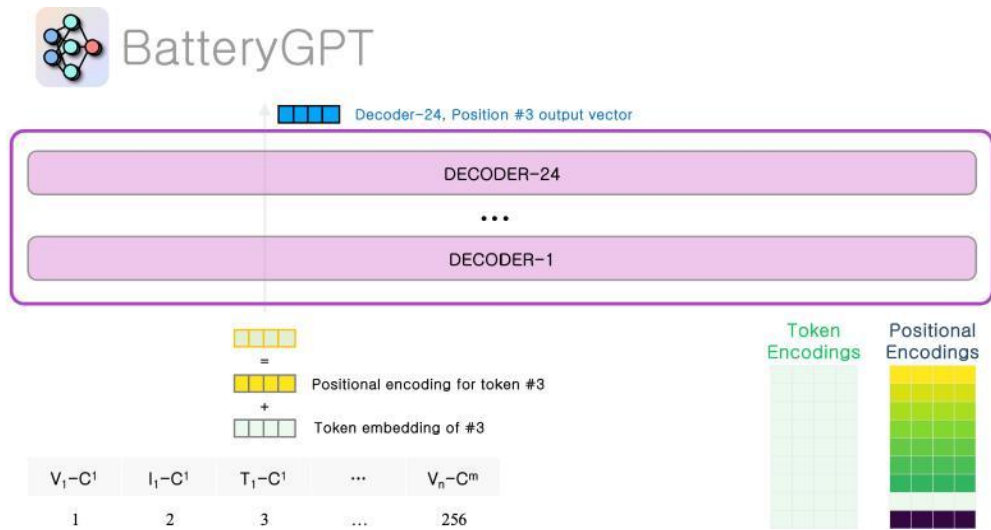
How to capture potential performance changes in future from early charging data is the key to achieve reliable early prediction of LIBs degradation. He et al.³⁴ utilize the first 100 cycles to generate complex image features as inputs to a CNN, demonstrating the capability of data-driven methods to uncover early degradation trends. Hsu et al.³⁵ directly infer the remaining useful life (RUL) solely from data in the first cycle, further reducing the early data requirements of data-driven approaches. Zhao et al.³⁶ utilize voltage, current, and temperature change curves extracted from the first 10 to 110 cycles to construct an input data structure tailored to a multi-branch Vision Transformer model, achieving relatively accurate early lifespan prediction using only 100 cycles of data. However, these methods rely on specific early cycles, leading to reduced prediction accuracy when critical data is missing or the distribution changes^{37,38}. Additionally, these methods require relatively complex feature engineering³⁹ and often lack intermediate feedback⁴⁰. These limitations pose substantial challenges for large-scale deployment, cross-platform application, and multi-chemistry battery lifetime prediction. To achieve efficient and universal early prediction of battery degradation, more flexible model structures and data-driven strategies must be explored.”

2. Clarified the Research Contributions

In the Methods section, we have reorganized and elaborated on the description of the BatteryGPT framework, explicitly explaining how its two-stage design addresses the identified research gaps.

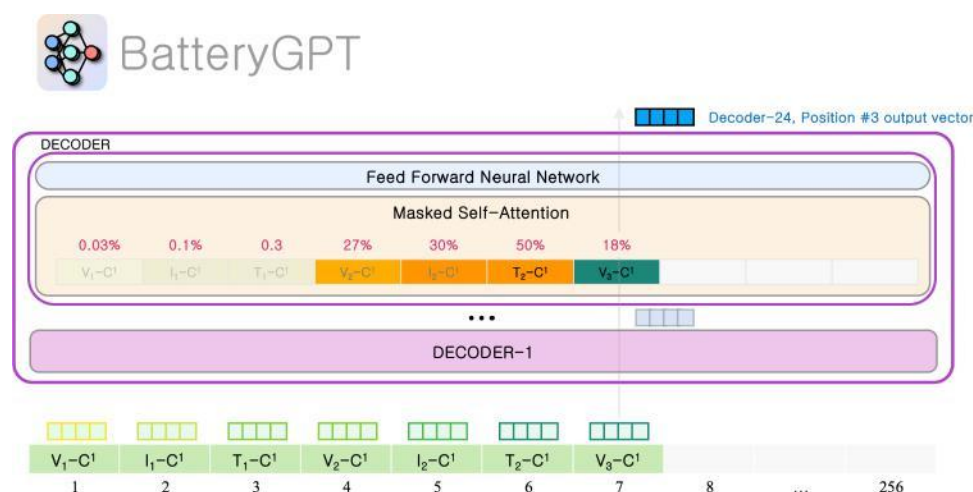
Revision in Methods section:

“The temporal dimension is implemented through the natural ordering of the continuous input sequence, working together with positional embedding to help the model decode the temporal and sequential structure of the input data without any feature engineering, as all parameters are learned directly from the data, as shown in Supplementary Fig. S1.”



Supplementary Fig. S1: The diagram shows the hierarchical structure of the 24-layer decoder stack, where each decoder layer processes input embeddings combined with positional encodings (Supplementary Fig. S2). The process starts with the token embedding of a voltage token (such as #1), which is then enhanced by adding the positional encoding for a specific token position (in this case, token #3). This combined input is processed sequentially through the decoders from 1 to 24 resulting in the output vector from the 24th decoder layer for the specified token position.

“The self-attention mechanisms and the autoregressive prediction process, which forms the foundation of our method to obtain the complete battery trajectory over all cycles, are shown in Supplementary Fig. S4.”



Supplementary Fig. S4: This diagram illustrates the detailed mechanism of masked selfattention within the decoder layer. It shows how different tokens in the input sequence, each represented by their token embeddings, are weighted during the attention process. The percentages indicate the attention weights assigned to various tokens (e.g., V_1-C^1 , I_1-C^1 , etc.) when processing the token at position #3 (T_3-C^1). These weights reflect the importance of each token in predicting the next token in the sequence. The output of the masked self-attention mechanism is then passed through a feed-forward neural network within the decoder layer. The diagram highlights the flow from the initial input embeddings to the final output vector from decoder-24 for token position #3.

“Hyperparameter Optimization of GPT-Small model is detailed in S.N5. In the BatteryGPT, the goal of the GPT-Small model is to predict the next token based on the current and previous tokens, enabling our method to start predictions from any cycle in the battery aging process. Therefore, a commonly used loss function is the cross-entropy loss. Specifically, we will use the model's output probability distribution to calculate the cross-entropy loss at each position and average over the entire sequence.”

3. Refined the Explanation of EPSO

We have revised the definition and purpose of EPSO to clearly state that EPSO represents the starting point for BatteryGPT’s prediction rather than the proportion of data used. The revised explanation further highlights that BatteryGPT’s input data volume is fixed at 256 tokens, corresponding to approximately 85 discharge voltage,

current, and temperature data points. This revision eliminates any confusion between EPSO and data usage volume, underscoring BatteryGPT's reduced data dependency and its advantage in flexible prediction starting points.

Revision in the Supplementary Materials:

“



Supplementary Figure S2: This diagram illustrates the structures of token encoding (wte) and positional encoding (wpe). The token encoding matrix (left) has dimensions of 763 (model vocabulary size) by 384 (embedding size), where each cell represents a specific token's embedding. The positional encoding matrix (right) has dimensions of 256 (context size) by 384 (embedding size), with each cell encoding positional information used to maintain the order of tokens within the context.



Supplementary Figure S5: Predictive starting points and data utilization of BatteryGPT under different EPSo settings. Taking a battery with a lifecycle of 600 cycles as an example. When EPSo = 0%, the input to the model consists of 256 tokens starting from the 1st cycle. When EPSo = 10%, the input to the model consists of 256 tokens starting from the 60th cycle. When EPSo = 30%, the input to the model consists of 256 tokens starting from the 180th cycle.

”

We sincerely appreciate your valuable feedback and have clearly indicated where these revisions have been incorporated into the manuscript for transparency and thoroughness. Thank you for helping us improve the rigor and clarity of our work, and we look forward to any further comments you may have.

References:

- [1] Saxena, S., Ward, L., Kubal, J., Lu, W., Babinec, S., & Paulson, N. (2022). A convolutional neural network model for battery capacity fade curve prediction using early life data. *Journal of Power Sources*, 542, 231736.
- [2] Liu, J., Thelen, A., Hu, C., & Yang, X. G. (2021, November). An end-to-end learning framework for early prediction of battery capacity trajectory. In Annual Conference of the PHM Society (Vol. 13, No. 1).

- 1131 Hong, J., Lee, D., Jeong, E. R., & Yi, Y. (2020). Towards the swift prediction of the remaining useful life of lithium-ion batteries with end-to-end deep learning. *Applied energy*, 278, 115646.
- 1141 Hsu, C. W., Xiong, R., Chen, N. Y., Li, J., & Tsou, N. T. (2022). Deep neural network battery life and voltage prediction by using data of one cycle only. *Applied Energy*, 306, 118134.
- 1151 He, N., Wang, Q., Lu, Z., Chai, Y., & Yang, F. (2024). Early prediction of battery lifetime based on graphical features and convolutional neural networks. *Applied Energy*, 353, 122048.
- 1161 Severson K A, Attia P M, Jin N, et al. Data-driven prediction of battery cycle life before capacity degradation11J1. *Nature Energy*, 2019, 4(5): 383-391.
- 1171 Khaleghi S, Hosen M S, Van Mierlo J, et al. Towards machine-learning driven prognostics and health management of Li-ion batteries. A comprehensive review11J1. *Renewable and Sustainable Energy Reviews*, 2024, 192: 114224.
- 1181 Wang, J., Zhang, C., Meng, X., Zhang, L., Li, X., & Zhang, W. (2024). A Novel Feature Engineering-Based SOH Estimation Method for Lithium-Ion Battery with Downgraded Laboratory Data. *Batteries*, 10(4), 139.
- 1191 Yang, M., Sun, X., Liu, R., Wang, L., Zhao, F., & Mei, X. (2024). Predict the lifetime of lithium-ion batteries using early cycles: A review. *Applied Energy*, 376, 124171.
- 11101 Yao, Z., Lum, Y., Johnston, A., Mejia-Mendoza, L. M., Zhou, X., Wen, Y., ... & Seh, Z. W. (2023). Machine learning for a sustainable energy future. *Nature Reviews Materials*, 8(3), 202215.
- 11111 Yang, A., Wang, Y., Tsui, K. L., & Zi, Y. (2019, May). Lithium-ion battery SOH estimation and fault diagnosis with missing data. In *2019 IEEE International Instrumentation and Measurement Technology Conference (I2MTC)* (pp. 1-6). IEEE.
- 11121 Demirci, O., Taskin, S., Schaltz, E., & Demirci, B. A. (2024). Review of battery state estimation methods for electric vehicles-Part II: SOH estimation. *Journal of Energy Storage*, 96, 112703.
- 11131 Pan, W., Luo, X., Zhu, M., Ye, J., Gong, L., & Qu, H. (2021). A health indicator extraction and optimization for capacity estimation of Li-ion battery using incremental capacity curves. *Journal of Energy Storage*, 42, 103072.
- 11141 Afshari, S. S., Cui, S., Xu, X., & Liang, X. (2021). Remaining useful life early prediction of batteries based on the differential voltage and differential capacity curves. *IEEE Transactions on Instrumentation and Measurement*, 71, 1-9.
- 11151 Fei, Z., Yang, F., Tsui, K. L., Li, L., & Zhang, Z. (2021). Early prediction of battery lifetime via a machine learning based framework. *Energy*, 225, 120205.
- 11161 Ma, Y., Li, J., Gao, J., & Chen, H. (2025). Prognostication of lithium-ion battery capacity fade based on data space compression visualization and SMA-ISVR. *Applied Energy*, 380, 124974.
- 11171 Mosallam, A., Medjaher, K., & Zerhouni, N. (2016). Data-driven prognostic method based on Bayesian approaches for direct remaining useful life prediction. *Journal of Intelligent Manufacturing*, 27, 1037-1048.
- 11181 Shah, A. A., Shah, N., Luo, L., Xing, W. W., Leung, P. K., Zhu, X., & Liao, Q. (2023). Nonlinear autoregressive models for high accuracy early prediction of Li-ion battery end-of-life. *Journal of Energy Storage*, 73, 109014.
- 11191 Wan, Z., Cheng, F., Mao, Z., Liu, Q., Song, H., Li, J., & Kurohashi, S. (2023). Gpt-re: In-

context learning for relation extraction using large language models. *arXiv preprint arXiv:2305.02105*.

[20]Aghajanyan, A., Yu, L., Conneau, A., Hsu, W. N., Hambardzumyan, K., Zhang, S., ... & Zettlemoyer, L. (2023, July). Scaling laws for generative mixed-modal language models. In International Conference on Machine Learning (pp. 265-279). PMLR.

[21]Afshari, S. S., Cui, S., Xu, X., & Liang, X. (2021). Remaining useful life early prediction of batteries based on the differential voltage and differential capacity curves. *IEEE Transactions on Instrumentation and Measurement*, 71, 1-9.

Reviewer #2

We express our sincere gratitude to the reviewer for providing valuable comments and suggestions. After careful modification of the manuscript, we are pleased to report that the quality of the paper has significantly improved. Your support is highly appreciated. The reply to the comments is as follows:

1. The authors have successfully addressed most of the issues raised in the previous review with clear and well-justified explanations. The manuscript now presents a more coherent argument, and the methodology is much better defined. However, I have a couple of remaining comments. While the manuscript cites relevant works, it lacks several key references in the transformer domain for battery prediction. Studies on transformers for early knee detection and SOC/SOH estimation are missing. Including these references and distinguishing how this work differs from them would significantly improve the paper's positioning within the state of the art. There are also some minor spelling and language issues throughout the manuscript. Although these do not detract significantly from the content, addressing them would enhance the overall readability and professionalism of the paper.

Authors' response:

Thank you for highlighting the importance of including additional references related to transformers in the context of battery prediction. We have revised the manuscript to incorporate several recent studies that apply transformer-based techniques to areas such as early knee detection and SOC/SOH estimation, and expanded the relevant literature reviews to underscore how our approach differs from these existing methods. While most transformer-based research employs purely supervised, end-to-end sequence-to-sequence or encoder–decoder frameworks [11, 2, 3, 4, 5], BatteryGPT leverages a GPT-style autoregressive mechanism within a two-stage framework [16]. This design allows for flexible prediction initiation (EPSO) and dynamic handling of missing early-cycle data, first generating an entire battery operational trajectory before refining health metrics (e.g., SOH) via a pre-trained estimator. By providing intermediate feedback rather than a strictly end-to-end prediction, BatteryGPT addresses known gaps in knee-point detection and offers transparent insights into evolving battery states. Furthermore, it minimizes reliance on manual feature extraction by operating directly on raw discharge voltage, current, and temperature data, a contrast to many transformer-based approaches (e.g. Vision-based Transformer with imaged features [14] and ICFormer with incremental capacity curves [15]) that depend on complex feature preprocessing. Through these clarifications, we highlight our unique contributions and position the manuscript more distinctly within the evolving literature on transformer-based methods for battery health management.

References:

- 111] Cai, Y., Li, W., Zahid, T., Zheng, C., Zhang, Q., & Xu, K. (2023). Early prediction of remaining useful life for lithium-ion batteries based on CEEMDAN-transformer-DNN hybrid model. *Heliyon*, 9(7).
- 112] Costa, N., Anseán, D., Dubarry, M., & Sánchez, L. (2024). ICFormer: A Deep Learning model for informed lithium-ion battery diagnosis and early knee detection. *Journal of Power Sources*, 592, 233910.
- 113] Zhao, S. Y., Ou, K., Gu, X. X., Dan, Z. M., Zhang, J. J., & Wang, Y. X. (2024). A novel

transformer-embedded lithium-ion battery model for joint estimation of state-of-charge and state-of-health. *Rare Metals*, 43(11), 5637-5651.

1141 Yang, A., Wang, Y., Tsui, K. L., & Zi, Y. (2019, May). Lithium-ion battery SOH estimation and fault diagnosis with missing data. In *2019 IEEE International Instrumentation and Measurement Technology Conference (I2MTC)* (pp. 1-6). IEEE.

1151 Costa, N., Anseán, D., Dubarry, M., & Sánchez, L. (2024). ICFormer: A Deep Learning model for informed lithium-ion battery diagnosis and early knee detection. *Journal of Power Sources*, 592, 233910.

1161 Yang, M., Sun, X., Liu, R., Wang, L., Zhao, F., & Mei, X. (2024). Predict the lifetime of lithium-ion batteries using early cycles: A review. *Applied Energy*, 376, 124171.

Revision in the Manuscript:

“

Existing data-driven solutions typically focus on a limited range of performance prediction, estimating LIB degradation over the next few or dozen charging cycles based on the latest charging data^{25,26}, demonstrating commendable accuracy in many applications²⁷. However, the effectiveness of data-driven methods largely depends on the quality and relevance of degradation features⁷. Severson et al.⁴ extract features from discharge voltage and capacity curves to predict battery life in early cycles, demonstrating the challenges of nonlinear performance degradation and feature analysis at the EOL for LIBs. Ma et al.²⁸ use battery capacity within a specific window as a feature to design a hybrid neural network combining convolutional neural network (CNN) and long short-term memory neural network (LSTM) to estimate capacity in the next window. Cai et al.²⁹ propose a CEEMDAN-Transformer-DNN hybrid model for RUL prediction, suggesting that future research should use various stress features, such as temperature, voltage, and current, as input parameters to achieve more advanced battery performance degradation prediction models due to the significant influence of actual operating conditions and working environments on capacity decay behaviour. Xu et al.³⁰ utilize Transformer as a global model to predict battery performance degradation, emphasizing the challenges of feature identification in early cycle data and long-term nonlinear prediction, clarifying that without additional corrections, Transformer cannot handle the severe nonlinear degradation at the EOL. Costa et al.³¹ develop ICFormer, a Transformer-based deep learning model that leverages self-attention mechanisms to analyse incremental capacity curves for early knee detection in lithium-ion batteries, highlighting the model's ability to identify degradation patterns and predict battery health indicators more effectively than traditional methods. Zhao et al.³² propose a novel Transformer-embedded model, combining Transformer-based feature extraction with enhanced unscented Kalman filter for SOC estimation, addressing challenges in aging feature identification under varying temperature and operating conditions. Therefore, early charging data often lacks sufficient degradation features, making it difficult for machine learning and deep learning models to recognize performance changes over the entire lifespan of LIBs, limiting their accuracy in predicting long-term degradation in the early stages³³.

How to capture potential performance changes in future from early charging data is the key to achieve reliable early prediction of LIBs degradation. He et al.³⁴ utilize the first 100 cycles to

generate complex image features as inputs to a CNN, demonstrating the capability of data-driven methods to uncover early degradation trends. Hsu et al.³⁵ directly infer the remaining useful life (RUL) solely from data in the first cycle, further reducing the early data requirements of data-driven approaches. Zhao et al.³⁶ utilize voltage, current, and temperature change curves extracted from the first 10 to 110 cycles to construct an input data structure tailored to a multi-branch Vision Transformer model, achieving relatively accurate early lifespan prediction using only 100 cycles of data. However, these methods rely on specific early cycles, leading to reduced prediction accuracy when critical data is missing or the distribution changes^{37,38}. Additionally, these methods require relatively complex feature engineering³⁹ and often lack intermediate feedback⁴⁰. These limitations pose substantial challenges for large-scale deployment, cross-platform application, and multi-chemistry battery lifetime prediction. To achieve efficient and universal early prediction of battery degradation, more flexible model structures and data-driven strategies must be explored. A feasible solution is to use early charging data to extrapolate future charging behaviours to predict SOH variations of the entire lifecycle. This requires the development of advanced deep learning models capable of handling long-term nonlinear changes and extracting subtle features from early charging data. Indeed, this is a potential application for causal language models. Causal language models⁴¹ have demonstrated powerful prediction and generation capabilities in Natural Language Processing (NLP). These models learn from vast amounts of text data, enabling to understand and predict the sequential structure of language⁴². Especially with increased model parameters and training data, they can adapt to various task scenarios. In the context of LIB degradation prediction, causal language models can capture and exploit long-term sequential dependencies in charging data, identifying subtle features from early charging cycles to infer and extend potential full lifecycle charging data⁴³. This approach may better understand and predict the long-term degradation of LIB performance, providing more accurate information for BMSs, thus enhancing LIB operational efficiency and safety.

”

Response to reviewers' comments follows here:

To Reviewer #1:

Thank you very much for your insightful comments and suggestions on our manuscript. In response, we have carefully revised the manuscript and addressed all points raised. Below we provide detailed responses to each comment and outline the corresponding changes made to the manuscript.

Thanks for addressing my comments. In regarding the advantages of your method, only MIT dataset is used in your case which can not fully validate or confirm your contributions compared with existing ones. For example, the data missing case.

Authors' response:

Thank you for your valuable feedback. We appreciate your thoughtful comments and suggestions.

R1. Why we used only the MIT dataset in the original submission

In the original submission we restricted evaluation to the MIT dataset published in Nature Energy because it is the community's primary benchmark for early-stage degradation prediction. That dataset uniquely provides consistent, high-resolution, cycle-resolved V/I/T trajectories together with an explicit linkage from the first tens of cycles to the subsequent end-of-life, which is exactly the regime our method targets. Many other public resources are valuable for different questions but are not directly aligned with our problem formulation or evaluation protocol: several mix heterogeneous cycling programs without a consistent mapping from early cycles to later failure [1], many lack clear and comparable annotations of the acceleration knee across cells [2], and a number do not provide full step-wise V/I/T at uniform resolution or focus primarily on calendar aging rather than long, uniform cycling [3]. Under these conditions, a tokenized, next-token generative model cannot be trained and assessed in a way that is comparable to the early-prediction literature, so MIT was the appropriate first testbed.

R2. New experiments beyond MIT: KIT dataset (2024–2025) with EV-like long-term aging

To address external validity and the reviewer's question about data conditions closer to field operation, we added a second evaluation on the KIT dataset (Scientific Data) [4], which contains long-horizon cycling under EV-like operating profiles with pronounced knee behavior. We used the same tokenization, the same early-prompt then free-running protocol, and chronological splits without leakage, and compared to carefully

implemented baselines. On KIT, BatteryGPT (EPSO = 5%) achieved RMSE 3.63%, MAPE 2.59%, knee-point error -62 cycles (KP MAPE 6.83%), and EOL error -12 cycles (EOL MAPE 1.23%). With a 30-cycle prompt, performance improved to RMSE 2.28%, MAPE 2.01%, knee-point error -34 cycles (KP MAPE 3.74%), and EOL error -3 cycles (EOL MAPE 0.37%). These results exceed the strongest regressive baseline (CNN-LSTM: RMSE 2.83%, MAPE 2.85%, knee-point error 33 cycles, KP MAPE 3.24%, EOL MAPE 3.25%) and stand in contrast to autoregressive LSTM/Transformer models run free-form over long horizons (RMSE 11.94–22.71%, MAPE 13.48–30.56%). The KIT experiments therefore corroborate that BatteryGPT maintains stable long-range generation and provides accurate SOH trajectories, knee timing, and EOL estimation beyond the original MIT benchmark; they also cover realistic logging artifacts, where the causal autoregressive formulation continues trajectories from partial prompts without a dedicated imputation module.

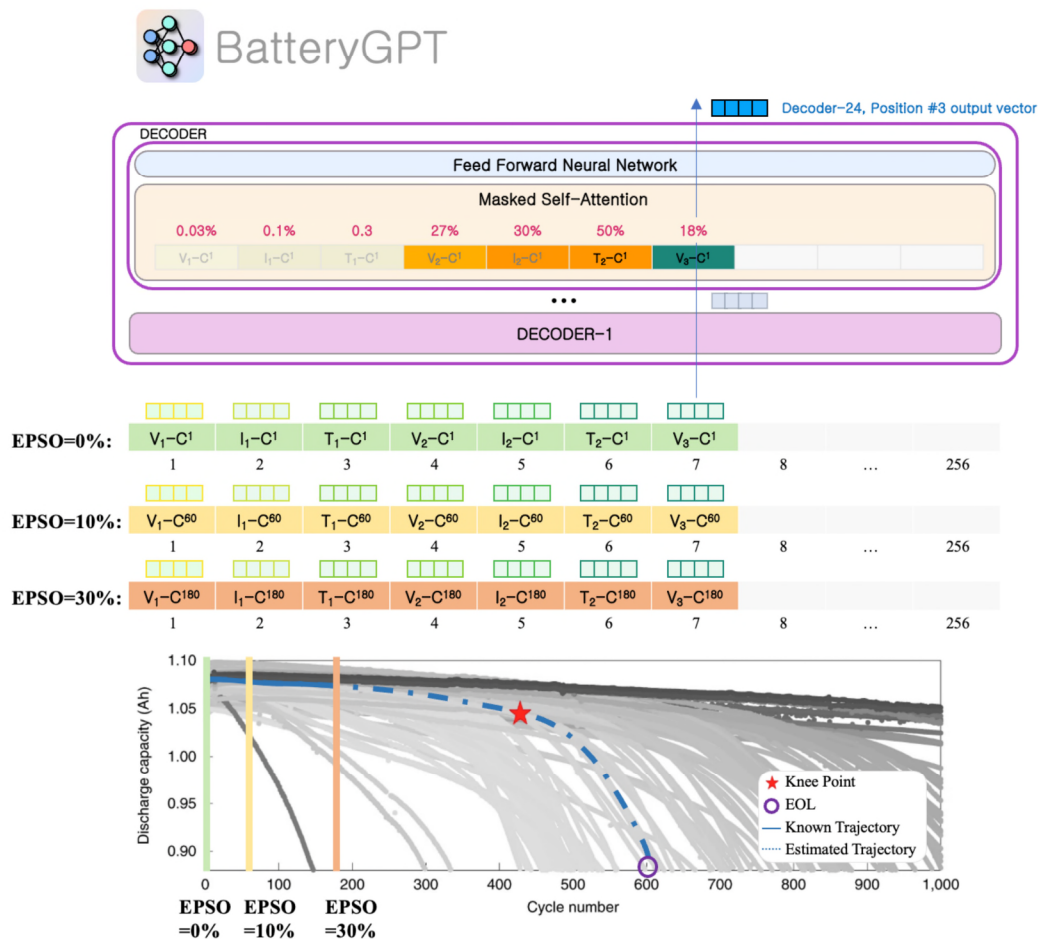
	Method	RMSE(%)	MAPE(%)	Knee Point Error	KP MAPE (%)	EOL Error	EOL MAPE (%)
Regressive prediction	CNN	7.41	7.27	98	9.62	Out of range	Out of range
	LSTM	12.41	15.58	-240	23.57	Out of range	Out of range
	CNN-LSTM	2.83	2.85	33	3.24	38	3.25
Autoregressive prediction	LSTM	11.94	13.48	-265	25.95	Out of range	Out of range
	Transformer	22.71	30.56	-421	41.07	Out of range	Out of range
	BatteryGPT(5)	3.63	2.59	-62	6.83	-12	1.23
	BatteryGPT(30)	2.28	2.01	-34	3.74	-3	0.37

Table 2. Comparison of LIB degradation prediction performance for KIT dataset. The out of range means that the predicted SOH do not reach 80%, and EOL can not be obtained.

R3. On “missing data”: why BatteryGPT is structurally suited to incomplete or partial trajectories

The concern about missing data is addressed by the way BatteryGPT conditions generation on the available left context. BatteryGPT is a causal autoregressive transformer over tokenized V/I/T sequences, trained to model the next token given only the preceding tokens. When part of a trajectory is absent, the model conditions on the remaining prompt and continues the sequence so that the generated suffix remains statistically consistent with the observed operating context. Supplementary Fig. S5 illustrates this behavior at the cycle scale via three “early-prediction starting offsets” (EPSO). With a fixed input length of 256 tokens, the model can be prompted from the beginning of life (EPSO = 0%), from 10% into life, or from 30% into life. In each case, the model generates the remainder of the life and the downstream SOH estimator recovers the knee point and EOL with comparable accuracy. This experiment is

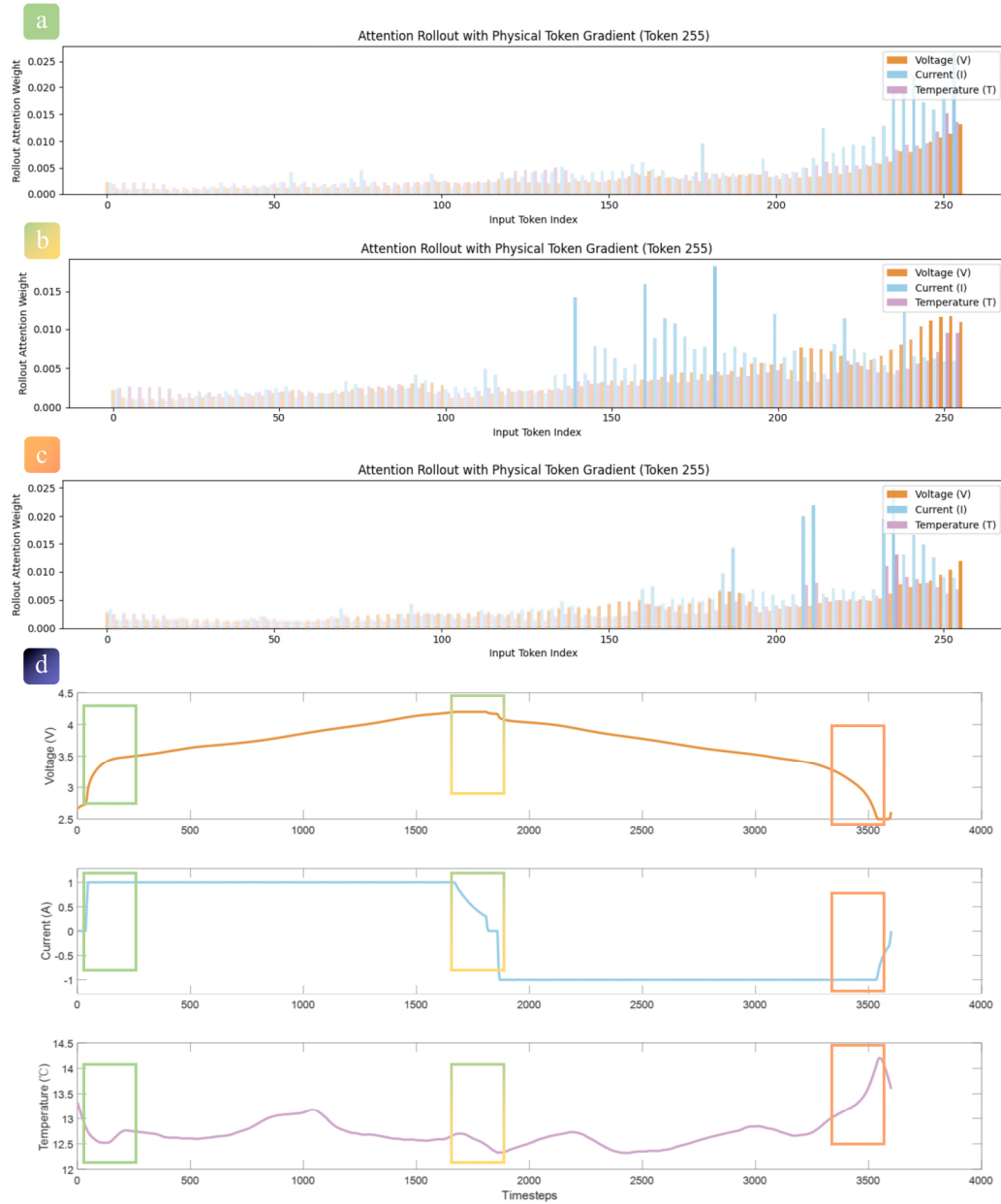
equivalent to having long missing prefixes and shows that the method does not require a separate imputation module to resume forecasting after gaps or late starts.



Supplementary Figure S5: Predictive starting points and data utilization of BatteryGPT under different EPSO settings. Taking a battery with a lifecycle of 600 cycles as an example. When EPSO = 0%, the input to the model consists of 256 tokens starting from the 1st cycle. When EPSO = 10%, the input to the model consists of 256 tokens starting from the 60th cycle. When EPSO = 30%, the input to the model consists of 256 tokens starting from the 180th cycle.

The attention analysis in Supplementary Fig. 6 further explains why the continuation is coherent. Panel d shows the physical V/I/T traces for one cycle with three colored context windows: an early charge-rise segment (green), a mid-cycle transition around current taper or voltage plateau (yellow), and a late end-of-discharge segment with a temperature excursion (orange). Panels a–c report the corresponding attention-rollout weights for the same prediction position (token 255) when the prompt is restricted to each colored window. Across all three contexts, the model allocates most weight to the most recent tokens, as expected under a causal mask, but it also places secondary mass on physically salient transitions: in the green window it emphasizes the initial voltage

rise and temperature settling; in the yellow window it emphasizes the current taper and plateau region; in the orange window it emphasizes the pre-EOD voltage curvature together with the temperature spike. These are precisely the features that precede subsequent deterioration and knee formation. The alignment between attention peaks and such operating signatures is observed on both MIT and KIT cells and remains visible when short segments are removed from the prompt, which explains why BatteryGPT can complete short gaps and then extrapolate reliably.



Supplementary Figure S6: Attention analysis of BatteryGPT (GPT-Small) under different context windows. (a–c) Gradient-weighted attention-rollout for the same prediction location (token 255) when the prompt is restricted to three distinct windows of

a single cycle; bars indicate the contribution of each modality (Voltage, Current, Temperature). (d) The corresponding physical V/I/T traces for the full cycle; the green, yellow, and orange boxes mark the early charge-rise segment, the mid-cycle taper/plateau region, and the late end-of-discharge segment with a temperature excursion, respectively—the exact contexts used in (a–c).

In practice, two use cases correspond to missing data. When there is a short interior gap, generation proceeds by conditioning on the available left context and continuing from the first token after the gap; when there is a long missing prefix, the EPSO protocol uses the first available window as the prompt and generates forward. In both cases, the downstream SOH head operates on short windows of generated tokens and produces a continuous SOH trajectory, from which knee timing and EOL are read. The revised manuscript reports that, under masking experiments that remove short segments from the prompt, trajectory errors and SOH errors remain within the same range as in complete-data runs, and that the KIT results reproduce the same behavior under EV-like cycling.

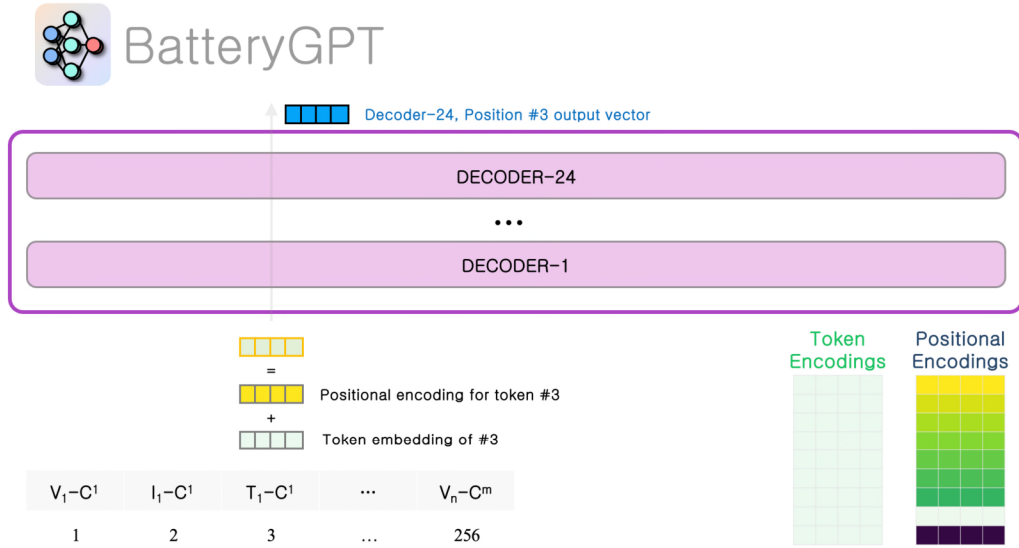
R4. Regarding the apparent lack of advantage over existing methods

We thank the reviewer for the thoughtful comment. We acknowledge that our previous presentation may not have sufficiently clarified the differences in task settings and data usage between our method and the baselines, which could have led to some misunderstanding regarding the performance comparison.

The core of our contribution lies in the design of a novel two-stage autoregressive prediction framework that enables long-term battery performance forecasting from early-life data. Under this framework, the model is expected to generate the entire degradation trajectory over the entire battery lifecycle using only a short segment of early-cycle data. We would like to clarify that existing autoregressive models such as LSTM and Transformer, even when trained under this same early-prediction setting, are unable to achieve comparable performance to our proposed method and cannot provide meaningful predictions for knee points and RUL.

In addition to this controlled comparison, we also recognize the need to contextualize our method’s performance with respect to commonly used baseline methods, the input data structure is shown in Supplementary Fig. 2. While these models typically rely on full lifecycle data and make predictions on a cycle-by-cycle basis, where the SOH of the next cycle is predicted based on the most recent full charge and discharge cycles (in this paper is the first three). Despite using less data (only early 30 percent of the lifecycle), BatteryGPT achieves superior or comparable accuracy in SOH, RUL, and knee point prediction as shown in Table. 1. This structure more clearly highlights the fact that our method performs better even under stricter data constraints.

This clearly demonstrates the advantage of BatteryGPT, which not only solves the more challenging early prediction problem, but also outperforms models that require more data and operate under more favorable conditions. We have revised the manuscript to make this distinction clearer and highlight these findings.



Supplementary Figure S1: The diagram shows the hierarchical structure of the 24-layer decoder stack, where each decoder layer processes input embeddings combined with positional encodings (Supplementary Fig. S2). The process starts with the token embedding of a voltage token (such as #1), which is then enhanced by adding the positional encoding for a specific token position (in this case, token #3). This combined input is processed sequentially through the decoders from 1 to 24, resulting in the output vector from the 24th decoder layer for the specified token position.

	Method	SOH Prediction RMSE(%)	SOH Prediction MAPE(%)	Knee Point Error	Knee Point MAPE(%)	RUL Error	RUL MAPE(%)
Regressive Prediction	CNN	6.33	5.88	-380	67.26	Out of Range	Out of Range
	LSTM	2.61	2.04	-149	26.37	-24	2.84
	CNN-LSTM	2.63	1.90	-20	3.54	-20	2.37
Autoregressive Prediction	LSTM	6.56	5.93	-491	86.9	Out of Range	Out of Range
	Transformer	6.49	5.73	-226	40.00	Out of Range	Out of Range
	BatteryGPT (5)	2.56	1.90	-177	31.33	-21	2.49
	BatteryGPT (30)	0.21	0.14	13	2.30	10	1.18

Table 1. Comparison of LIB degradation prediction performance. The out of range means that the predicted SOH do not reach 80%, and EOL can not be obtained.

In the previous version of the manuscript, Table 1 also included the result of BatteryGPT (5). However, during the revision process, we realized that this version of

the model had already been thoroughly discussed in Fig. 3 and the corresponding analysis, which demonstrates how the performance of BatteryGPT improves as the EPSO (Early Prediction Starting Offset) increases. Including BatteryGPT (5) again in Table 1 would therefore be redundant and potentially confusing, as the table is intended to emphasize the performance of the proposed method under the same early prediction setting. To streamline the presentation and avoid repetition, we have removed the BatteryGPT (5) entry from Table 1 in the revised manuscript, while ensuring that its performance trend remains clearly presented and discussed in the figure and main text.

The compared work [5] uses the same MIT battery dataset consisting of 124 lithium iron phosphate/graphite cells (1.1 Ah nominal capacity). For each charge cycle, continuous measurements of voltage, current, temperature, and charging capacity were collected at fixed time intervals during real-world charging procedures. These time-series data were then preprocessed into a quasi-video format and used as inputs to a conditional variational autoencoder (RCVAE), with EOL (End of Life) and ECL (Equivalent Cycle Life) serving as conditional labels. Regarding capacity prediction, the RCVAE model achieved an RMSE of 0.038–0.044 Ah on the MIT dataset. Given the nominal battery capacity of 1.1 Ah, this corresponds to an SOH prediction RMSE of approximately **3.45%**.

The paper [6] employs charging voltage, current, and temperature time-series data as input, where the first 1000 time steps of each charging cycle are used and represented as a 1000×3 matrix. The method is trained and evaluated using NASA dataset, which containing full lifespan cycling data of 18650 lithium-ion batteries. On the NASA dataset, the proposed DSTCN model achieved an SOH prediction RMSE of **1.47%**. It is also important to note that the DSTCN-based method relies solely on the first 1000 time steps of the current charging cycle to estimate the SOH, which means it cannot perform early prediction of SOH before degradation becomes evident. Therefore, its function is more appropriately categorized as SOH estimation rather than true SOH prediction. This limits its practical applicability in scenarios that require early-warning diagnostics or remaining useful life forecasting.

The paper [7] proposes a domain-adaptive deep learning framework that estimates battery SOH without requiring aging data from the target battery. The model takes as input raw time-series data of voltage, current, temperature, and time from individual charge/discharge cycles, and outputs the corresponding SOH. It is trained using a large dataset comprising 71,588 cycles collected from 65 commercial cells across five manufacturers. The key focus of this work is cross-domain generalization. In each experiment, the model is trained on data from four manufacturers (source domain) and evaluated on batteries from the remaining manufacturer (target domain), without using

SOH labels from the target domain. This setup emphasizes robustness under domain shift, rather than learning from limited early-cycle data or performing early-stage SOH forecasting. The method achieves strong SOH estimation results, with absolute errors $\leq 3\%$ for 89.4% of test samples and $\leq 5\%$ for 98.9%.

The paper [8] introduces a deep learning framework that leverages multi-modal sensor data collected from large-scale vehicle fleets. Inputs include voltage, current, temperature, SOC, and other real-world operating parameters obtained from BMS and on-board sensors. The model is trained to estimate battery SOH based solely on historical in-vehicle usage data, without requiring any laboratory aging tests. Their model achieves a **mean absolute percentage error (MAPE) of 2.83%**. However, it is built for large-scale fleet monitoring using mid-life historical data and does not support early-cycle prediction or data-efficient inference.

The paper [9] employs early-life discharge voltage curves as input features, extracted from the first 5 to 100 charge–discharge cycles of lithium-ion cells before significant degradation occurs. The method is trained and evaluated using a large dataset of commercial cells cycled under various conditions, with the objective of predicting total cycle life from only this early-cycle data. A regularized linear model is applied to engineered voltage-time features to forecast the battery’s end-of-life cycle count. On the held-out test set, the model achieved a mean absolute percentage error of 9.1% for quantitative life prediction and a classification error rate of 4.9% for high-/low-life categorization using only the first 5 cycles. While this work demonstrates the feasibility of long-term life prediction from minimal early data, it does not directly output SOH trajectories, knee points, or remaining useful life in real time, and is therefore focused specifically on lifetime forecasting rather than comprehensive SOH prediction.

The paper [10] proposes a model-free framework for reconstructing the full capacity degradation trajectory of lithium-ion batteries using only early-cycle data. The method takes charge–discharge voltage and capacity measurements from as few as the first three cycles and predicts several fixed interpolation points along the future SOH curve, which are then connected to generate an approximate degradation profile. The approach is evaluated on a dataset of 169 commercial cells cycled under various conditions, achieving a mean absolute percentage error below 1.60% in trajectory prediction under five-fold cross-validation. It is important to note that this method can only forecast SOH at those fixed interpolation points and relies on interpolation to estimate the rest of the curve, rather than directly predicting the SOH for every cycle across the full lifetime as in BatteryGPT. While the method demonstrates strong capability in long-term forecasting from minimal early data, it does not explicitly provide standalone EOL or knee-point predictions.

Method	Early-cycle data prediction	Requires full-lifecycle data	SOH prediction	SOH estimation	RUL/EOL prediction	Knee point prediction
[5]	X	✓	✓	X	X	X
[6]	X	✓	X	✓	X	X
[7]	X	✓	X	✓	X	X
[8]	X	✓	X	✓	X	X
[9]	✓	X	X	X	✓	X
[10]	✓	X	✓	X	X	X
BatteryGPT	✓	X	✓	✓	✓	✓

In summary, the above works address different aspects of SOH estimation. In contrast, our method targets accurate SOH prediction using early-cycle, limited data, and demonstrates a significantly lower RMSE of **0.21%**, reflecting superior accuracy and broader applicability in scenarios requiring early diagnostics or life forecasting.

[1] Preger, Y., Barkholtz, H. M., Fresquez, A., Campbell, D. L., Juba, B. W., Romàn-Kustas, J., ... & Chalamala, B. (2020). Degradation of commercial lithium-ion cells as a function of chemistry and cycling conditions. *Journal of The Electrochemical Society*, 167(12), 120532.

[2] Sahaand, B., & Goebel, K. (2007). Battery Data Set, NASA ames prognostics data repository. NASA Ames Research Center.

[3] Stroebel, F., Petersohn, R., Schricker, B., Schaeufl, F., Bohlen, O., & Palm, H. (2024). A multi-stage lithium-ion battery aging dataset using various experimental design methodologies. *Scientific data*, 11(1), 1020.

[4] Luh, M., & Blank, T. (2024). Comprehensive battery aging dataset: capacity and impedance fade measurements of a lithium-ion NMC/C-SiO cell. *Scientific Data*, 11(1), 1004.

[5] Jiang, L., Hu, C., Ji, S., Zhao, H., Chen, J., & He, G. (2025). Generating comprehensive lithium battery charging data with generative AI. *Applied Energy*, 377, 124604.

- [6] Tang, Y., Yang, K., Zheng, H., Zhang, S., & Zhang, Z. (2022). Early prediction of lithium-ion battery lifetime via a hybrid deep learning model. *Measurement*, 199, 111530.
- [7] Lu, J., Xiong, R., Tian, J., Wang, C., & Sun, F. (2023). Deep learning to estimate lithium-ion battery state of health without additional degradation experiments. *Nature Communications*, 14(1), 2760.
- [8] Liu, H., Li, C., Hu, X., Li, J., Zhang, K., Xie, Y., ... & Song, Z. (2025). Multi-modal framework for battery state of health evaluation using open-source electric vehicle data. *Nature Communications*, 16(1), 1137.
- [9] Severson, K. A., Attia, P. M., Jin, N., Perkins, N., Jiang, B., Yang, Z., ... & Braatz, R. D. (2019). Data-driven prediction of battery cycle life before capacity degradation. *Nature Energy*, 4(5), 383-391.
- [10] Kim, S., Jung, H., Lee, M., Choi, Y. Y., & Choi, J. I. (2023). Model-free reconstruction of capacity degradation trajectory of lithium-ion batteries using early cycle data. *ETransportation*, 17, 100243.

Revision in the Manuscript:

“In addition, the KIT dataset⁴⁶, which exhibits similar degradation characteristics and includes clearly observable knee points and EOL are used to further validate the generalization capability of the proposed method. The experimental results on the KIT dataset are presented in (S.N6), demonstrating consistent predictive performance and supporting the robustness of our approach.”

“Table. 1 presents evaluation results when EPSO is set to 30% (BatteryGPT (30)), showcasing impressive early-stage prediction performance of BatteryGPT over the baseline models. In SOH prediction, BatteryGPT (30) achieves an RMSE of just 0.21% and a MAPE of 0.14%, significantly leading all baselines. Moreover, BatteryGPT (30) also excels in predicting knee point and EOL, with errors of only 13 cycles and 10 cycles and MAPEs of 2.30% and 1.18%, while existing autoregressive baselines such as LSTM and Transformer are unable to provide meaningful predictions for knee points and RUL.

Furthermore, even when compared with regressive baselines that leverage full-cycle data during the entire battery life, BatteryGPT (30) still shows superior performance in all metrics. This comparison highlights the strong generalization and long-term prediction capability of BatteryGPT, even when operating under stricter data constraints. These results clearly demonstrate the practical advantages of the proposed two-stage

autoregressive prediction framework.”

Revision in the Supplementary:

“

Supplementary Note 6: Evaluation on the KIT Dataset

The KIT dataset is introduced to assess the external validity of BatteryGPT beyond the MIT benchmark used in the main text. KIT contains long-horizon cycling under EV-like operating profiles, multiple ambient temperatures, and pronounced accelerated-degradation knees. This note summarizes the evaluation protocol, baselines, metrics, and results on KIT, and explains why they substantiate the claims made for BatteryGPT.

	Method	RMSE(%)	MAPE(%)	Knee Point Error	KP MAPE (%)	EOL Error	EOL MAPE (%)
Regressive prediction	CNN	7.41	7.27	98	9.62	Out of range	Out of range
	LSTM	12.41	15.58	-240	23.57	Out of range	Out of range
	CNN-LSTM	2.83	2.85	33	3.24	38	3.25
Autoregressive prediction	LSTM	11.94	13.48	-265	25.95	Out of range	Out of range
	Transformer	22.71	30.56	-421	41.07	Out of range	Out of range
	BatteryGPT(5)	3.63	2.59	-62	6.83	-12	1.23
	BatteryGPT(30)	2.28	2.01	-34	3.74	-3	0.37

Table 2. Comparison of LIB degradation prediction performance for KIT dataset. The out of range means that the predicted SOH do not reach 80%, and EOL can not be obtained.

Dataset and preprocessing.

Cells are cycled over long durations with periodic check-ups and temperature variation. We adopt the same tokenization scheme used for MIT: voltage, current, and temperature are discretized into non-overlapping vocabularies with fixed offsets; per-dataset numeric ranges are estimated from the training split with a small safety buffer. All experiments use chronological splits. The SOH estimator operates on short windows of generated tokens, identical to the design described in the main text.

Evaluation protocol.

BatteryGPT is prompted with the initial portion of a life and then run in a free-running, fully autoregressive mode to the end of life. We report two prompt lengths: BatteryGPT(5) and BatteryGPT(30), where the number denotes the number of observed cycles provided as the prompt before free generation. This protocol matches the early-prediction use case. For interpretability and the missing-data analysis, we also consider

predictive starting offsets (EPSO) as illustrated in Supplementary Fig. S5: with a fixed prompt length (256 tokens), the model can start at 0%, 10%, or 30% of life and generate the remainder; this is equivalent to late starts or long missing prefixes.

Baselines.

We compare against (i) regressive predictors trained to map early cycles directly to SOH/RUL: CNN, LSTM, and CNN-LSTM; and (ii) naïve autoregressive forecasters (LSTM and Transformer) run free-form over long horizons. All baselines use the same splits and early-prompt regime; hyperparameters are tuned on validation data.

Metrics.

We report (i) SOH RMSE (%) and MAPE (%); (ii) Knee-Point (KP) Error in cycles and KP MAPE (%); and (iii) EOL Error in cycles and EOL MAPE (%). Knee and EOL are computed from the predicted SOH trajectory using the same rules across all methods.

Results.

On KIT, BatteryGPT achieves accurate long-horizon trajectory prediction and downstream prognostics:

- BatteryGPT(5) (EPSO = 5 cycles): RMSE 3.63%, MAPE 2.59%, KP Error −62 cycles, KP MAPE 6.83%, EOL Error −12 cycles, EOL MAPE 1.23%.
- BatteryGPT(30) (EPSO= 30 cycles): RMSE 2.28%, MAPE 2.01%, KP Error −34 cycles, KP MAPE 3.74%, EOL Error −3 cycles, EOL MAPE 0.37%.

These results compare favorably with the strongest regressive baseline CNN-LSTM (RMSE 2.83%, MAPE 2.85%, KP Error 33 cycles, KP MAPE 3.24%, EOL MAPE 3.25%) and are markedly better than naïve long-horizon autoregressive LSTM/Transformer forecasters (RMSE 11.94–22.71%, MAPE 13.48–30.56%), which accumulate exposure bias when run free-form.

”

Response to reviewers' comments follows here:

To Reviewer #2:

Thank you very much for your insightful comments and suggestions on our manuscript. In response, we have carefully revised the manuscript and addressed all points raised. Below we provide detailed responses to each comment and outline the corresponding changes made to the manuscript.

The code looks good and can be run, although they do not provide any weights for the model, so it is not straightforward to obtain the same results as in the article.

Authors' response:

Thank you for pointing out the reproducibility issue. We would like to clarify that pretrained checkpoints are already provided with the submission. The repository (<https://github.com/ReparkHjc/BatteryGPT>) README contains a "Pretrained models" section with direct download instructions for the BatteryGPT checkpoints and the SOH estimator. The evaluation scripts in the repo load these weights by default and reproduce the tables and figures in the paper using the specified data splits and seeds.

For long-term accessibility, we are archiving the checkpoints and configs on a DOI-backed repository and mirroring them from the project page. These changes remove any ambiguity about which weights to use and allow a one-click reproduction of the reported results.